

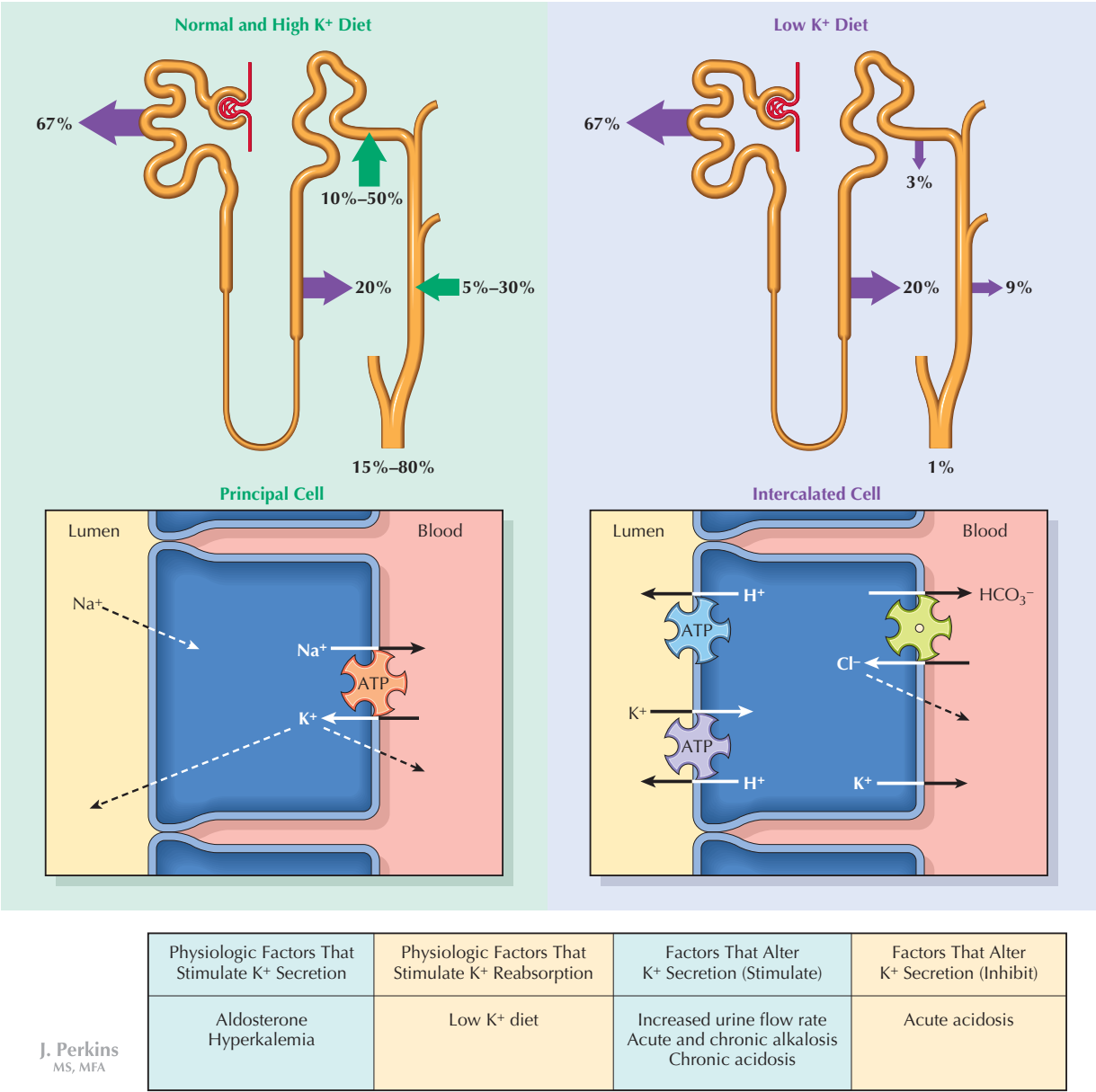


Figure 17.4 Renal Potassium Handling To maintain normal plasma K⁺ concentration (3.5 to 5 mEq/L), the kidney must control K⁺ excretion, and the amount of K⁺ excreted changes with dietary intake. Diets low in K⁺ stimulate avid K⁺ reabsorption throughout the nephron, whereas diets high in K⁺ stimulate distal K⁺ secretion (in green).

their rate of reabsorption. Most of the calcium and phosphate in the body (99% and 85%, respectively) is found in bone matrix. Renal phosphate and calcium reabsorption are both regulated by PTH (see Chapter 30).

Calcium Handling

About 40% of plasma Ca²⁺ is bound to proteins, leaving 60% free for filtration at the glomeruli. The kidneys reabsorb

~99% of the filtered Ca²⁺ at sites throughout the nephron (Fig. 17.5A):

- *Proximal tubule:* Ca²⁺ reabsorption is paracellular, via solvent drag initiated by bulk reabsorption of Na⁺ and water. This accounts for ~70% of Ca²⁺ reabsorption.
- *Thick ascending limb of Henle (TALH):* Reabsorption is paracellular, again in parallel with Na⁺ reabsorption. In addition, the lumen-positive transepithelial potential

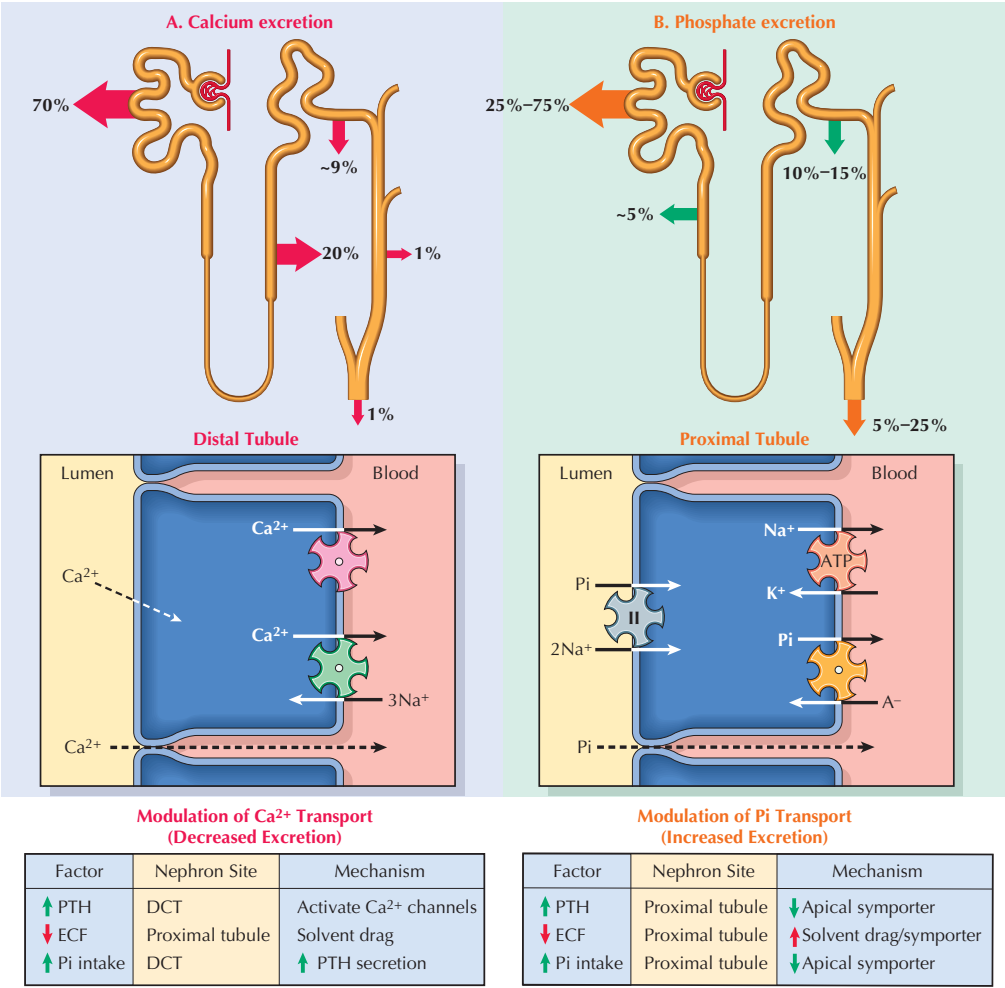


Figure 17.5 Renal Calcium and Phosphate Handling Calcium is reabsorbed along much of the nephron, and very little is excreted. Regulation of distal calcium reabsorption is by parathyroid hormone (PTH), which opens apical calcium channels. Under normal conditions, ~75% of the filtered load of phosphate is reabsorbed, with all of the reabsorption occurring in the proximal tubule via Na^{+} - Pi cotransporters. This is highly dependent on the dietary intake of phosphate as well as PTH levels. In response to PTH, proximal tubular reabsorption of phosphate is inhibited, and phosphate excretion increases. This also occurs with diets high in phosphate. Low-phosphate diets significantly increase Pi reabsorption, recruiting transporters in sites distal to the proximal convoluted tubule (in green), which can reduce phosphate excretion to 5% to 10%.

favors paracellular reabsorption of divalent cations in this segment (~20% of reabsorption). Because Ca^{2+} follows sodium reabsorption, changes in sodium reabsorption (such as with loop diuretics) will also reduce Ca^{2+} reabsorption.

- **Distal tubule:** Although this segment accounts for only ~8% to 9% of Ca^{2+} reabsorption, this is the site of hormonal control. Transport is transcellular and is facilitated by the high electrochemical gradient from the tubule into the cell. Once in the cell, transport into the interstitium is through active Ca^{2+} ATPase and $\text{Na}^{+}/\text{Ca}^{2+}$ exchangers on the basolateral membrane (see Fig. 17.5A). The transporters in the distal tubule are

under hormonal control by parathyroid hormone (PTH).

Renal handling of Ca^{2+} is regulated by the effects of PTH on calcium transporters. Low plasma Ca^{2+} directly stimulates PTH release from the parathyroid glands (see Chapter 30). PTH activates apical calcium channels and stimulates basolateral Ca^{2+} transporters in the distal tubule.

Phosphate Handling

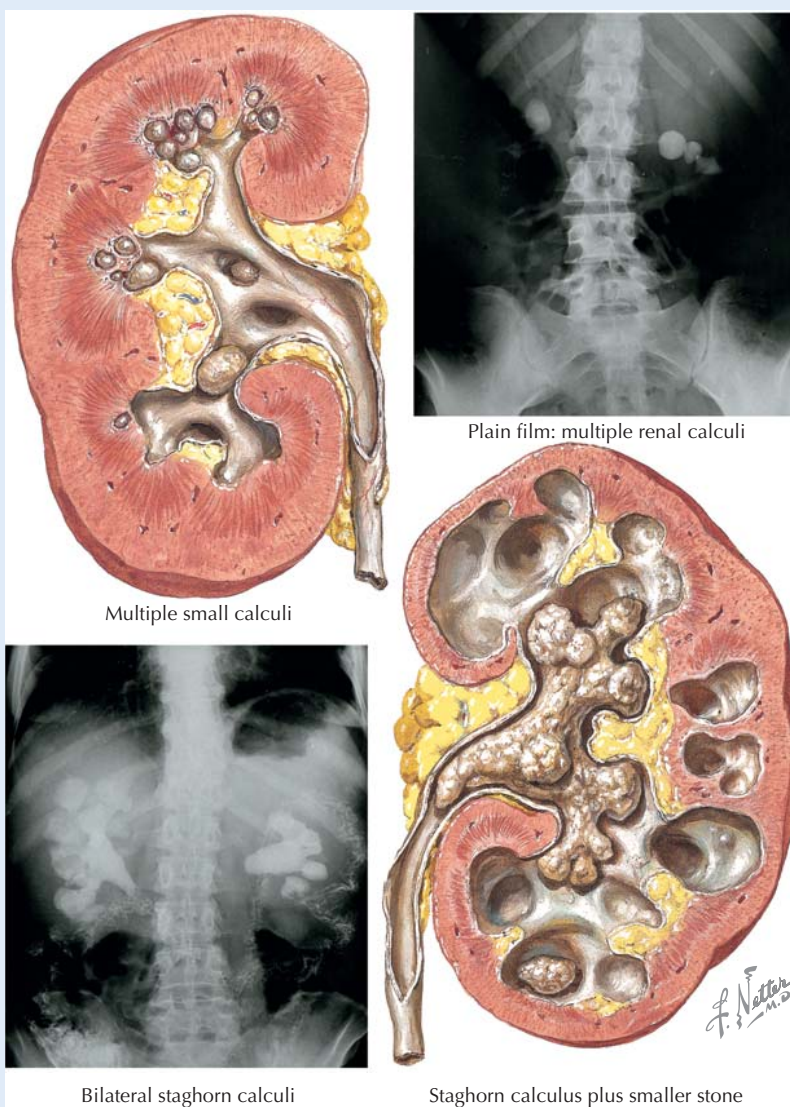
Phosphates are required for bone matrix formation as well as for intracellular high-energy mechanisms (e.g., ATP

CLINICAL CORRELATE

Kidney Stones (Renal Calculi)

Kidney stones are solid aggregates of minerals that form in the kidney (nephrolithiasis) or ureters (urolithiasis). The size of stones is variable, and many small stones will pass through the ureters and urethra without problem. However, if stones grow large enough (2 to 3 mm), they can block the ureter and cause intense

pain and vomiting. The most common stones are calcium oxalate, and it is the presence of oxalate (not the calcium) that drives mineral precipitation. Treatment depends on size of the stone and duration of the blockage. Typically, unless there are severe symptoms, small stones will be left to pass; however, long-term (more than 30 days) blockage can result in renal failure, and intervention with stent placement and laser or ultrasound may be performed.



Renal Calculi

formation and utilization). The majority of plasma phosphate (Pi) (more than 90%) is available for filtration, and Pi reabsorption and excretion are highly dependent on diet and age. As with glucose, Pi has a TM that can be saturated. Under normal dietary conditions, transporters are present only in

the proximal tubules, and ~75% of the filtered phosphate is reabsorbed by apical Na^+ -Pi cotransporters (see Fig. 17.5B). The remaining 25% of the Pi load is excreted; part of the Pi can be used to buffer H^+ , forming titratable acids (see Chapter 20).

In growing children and with diets low in Pi, Na⁺-Pi cotransporters are also present in the proximal straight tubules and distal tubules, facilitating reabsorption of up to 90% of the filtered Pi load.

Renal Pi reabsorption is primarily controlled by diet and parathyroid hormone (PTH), both of which affect the number of Na⁺-Pi cotransporters in the apical membranes:

- **Diet:** High dietary Pi causes reduction in the number of Na⁺-Pi cotransporters in the apical membrane, increasing Pi excretion. Conversely, low dietary Pi will increase transporters on the proximal tubule brush border, as well as in sites distal to the proximal tubule. This will allow avid Pi uptake and reduced urinary Pi excretion.

- **PTH:** PTH is secreted from the parathyroid glands in response to high plasma Pi concentrations or low plasma calcium concentrations. It decreases apical Na⁺-Pi cotransporters, reducing reabsorption and increasing urinary excretion of Pi.

Plasma Ca²⁺ and phosphate regulation are intertwined because of the constant bone resorption and deposition. In response to low plasma Ca²⁺, vitamin D increases intestinal calcium and phosphate absorption, and PTH induces bone resorption—both actions increase Ca²⁺ and phosphate in ECF. At the kidneys, PTH increases calcium reabsorption but compensates for the additional ECF phosphate by decreasing phosphate transporters and increasing urinary phosphate excretion. Thus, by this mechanism the kidneys regulate extracellular Ca²⁺ and phosphate concentrations.

CLINICAL CORRELATE

Hyponatremia

Hyponatremia is defined as the state of low plasma sodium (less than 135 mEq/L). This can be caused by several mechanisms that result in low sodium concentrations and reduced plasma osmolarity. During hyponatremia, fluid shifts *into* cells, reestablishing normal ECF osmolarity but causing cellular swelling. This can have important effects, especially on brain tissues which are confined to a bony space and are unable to tolerate swelling:

- Rapid fluid shifts into cells can be a critical problem, because acute cerebral swelling can lead to disoriented mental status, seizures, coma, and death. In these cases, reducing the ECF is necessary to draw the fluid out of cells. Water restriction and/or ADH (V1) antagonists are used to increase urinary free water excretion.
- If the hyponatremia is established over time (for example, in Addison's disease), brain tissues compensate for fluid shifts by decreasing intracellular content of osmolytes (organic solutes such as inositol and glutamine). This reduces the osmotic force that would draw fluid into the cells and allows the cells to maintain normal volume. Because of this, treatment of hyponatremia should involve slow restoration of salt and fluid balance to normal levels. Otherwise, the brain cells will shrink, inducing an acute, potentially critical, intracellular imbalance. Gradual correction of this type of hyponatremia will allow the osmolytes to increase in brain cells.

Exercise-associated hyponatremia (EAH) can occur as a result of fluid and electrolyte losses through sweat during long-term exercise (marathons, triathlons). Although most people do not experience a serious drop in ECF Na⁺ concentration, the critical cases of EAH are most likely to occur from a combination of the following:

- An initial imbalance of fluid and electrolyte losses, due to overhydration during the exercise.

- Acute syndrome of inappropriate ADH secretion (SIADH) that can occur because of *large* fluid losses. Recall that both increased osmolarity and fluid losses can stimulate ADH, but that the system is more sensitive to changes in ECF osmolarity than to changes in fluid volume. However, if the volume loss continues and becomes severe, when a dehydrated athlete drinks too much hypotonic fluid, the increase in ADH will cause excessive free water reabsorption by the collecting ducts, rapidly decreasing the ECF sodium concentration. In this scenario, the need to compensate for the volume will override the sodium levels, ADH secretion will continue, and plasma Na⁺ can fall to critically low levels (below 125 mEq/L).

Early symptoms include bloating, nausea, vomiting, and headaches, which can progress to disorientation, seizures, and death if not immediately treated. Hyponatremia can be prevented by restricting water intake (drinking only when thirsty). Although overhydration during the exercise is a direct cause of hyponatremia, risk factors for developing EAH include low body weight, female sex, and inexperience with marathons. ADH V2 receptor antagonists are used to treat severe hyponatremia.

Postsurgical acute hyponatremia frequently occurs in elderly patients. The stress of surgery can cause an acute SIADH, rapidly increasing free water reabsorption and reducing ECF Na⁺ concentration. As stated earlier, treatment should begin immediately, with water restriction (to limit further fluid retention) and a V2 antagonist. Correcting the hyponatremia restores normal function.

Pseudohyponatremia occurs when there is an incorrect, low measurement of plasma sodium due to conditions that produce high lipid or proteins (e.g., hyperlipidemia, hyperproteinemia) in the blood. In this case, the substances reduce the total plasma fluid and although the amount of sodium is normal, the clinical measurement may be falsely low. Thus, the person is not hyponatremic and treatment will focus on reducing the lipids and/or proteins.

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CHAPTER 18

Urine Concentration and Dilution Mechanisms

THE LOOP OF HENLE AND COLLECTING DUCT CELLS

Ultimately, the regulation of plasma osmolarity and volume are the responsibility of the loop of Henle and collecting ducts (CDs) and the vasa recta. Changes in the permeability of the loop of Henle to solutes and water allow for the concentration and dilution of the tubular fluid, as well as the ability of the kidneys to regulate overall water and solute reabsorption. Reabsorption is facilitated by the vasa recta that surround the medullary tubules and collecting ducts.

The descending and ascending limbs of the loops of Henle have specific permeability characteristics:

- *Descending limbs of Henle's loop are concentrating segments:* permeable to water, impermeable to reabsorption of solutes (urea can be secreted into the tubule, further concentrating the tubular fluid).
- *Thick ascending limbs of Henle's loop are diluting segments:* impermeable to water, but $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ transporters reabsorb electrolytes, thus diluting the tubular fluid.

With this mechanism in place, the tubular fluid entering the distal tubule has an osmolarity of ~ 100 mosm/L; the fine-tuning to concentrate urine will occur in the collecting ducts. There are **antidiuretic hormone (ADH)**–sensitive water channels in the collecting duct cells that allow solute-free water reabsorption and concentration of the hypo-osmotic tubular fluid. However, this concentration can only be achieved if an osmotic gradient exists from tubular lumen to the interstitial space.

URINE CONCENTRATING MECHANISM

The Medullary Interstitium

The ability to reabsorb solute-free water in both the descending limb of Henle and the collecting ducts is possible because of the osmolar concentration gradient within the medullary interstitial fluid and the presence of specific water channels in the collecting duct cells. Water can only move when there is an osmotic gradient; the factors contributing to the water

movement are illustrated by the numbers in red in Figure 18.1. In this interstitial gradient, the osmolarity is ~ 300 mosm/L at the corticomedullary border and rises to ~ 1200 mosm/L in the deepest part of the medulla. With this gradient in place, if water channels are present, water from the tubules readily diffuses into the interstitium (with its higher osmolar concentration), and then into the vasa recta network.

Medullary Countercurrent Multiplier

The interstitial osmolar gradient from cortex to inner medulla is formed and maintained by the coordinated efforts of the ascending and descending limbs of Henle and their selective permeability to solutes. Important factors contributing to establishing and maintaining the gradient are the $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ transporters in the *thick ascending limb of Henle (TALH)*, water absorption in the descending limb of Henle, the constant flow of tubular fluid through the loops, and urea recycling (urea reabsorption from the collecting ducts and urea diffusion from the medullary interstitium into the thin limb of Henle). Creation of the gradient begins with the following:

- The $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ transporters in the TALH, which transport solutes into the interstitium, increasing interstitial fluid osmolarity and decreasing tubular fluid osmolarity.
- This *increased interstitial osmolarity promotes free water reabsorption from the descending limb of Henle*, which increases tubular fluid osmolarity in the descending limb (concentrating limb).
- As tubular fluid flows from the descending limb, the more concentrated tubular fluid flows into the TALH and the solutes are transported into the interstitium through $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ transporters, further increasing interstitial fluid osmolarity. The higher interstitial osmolarity facilitates more free water reabsorption from the descending limb, concentrating the tubular fluid. This cycle (movement of concentrated tubular fluid into the TALH, transport of solutes by the $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ transporters into the interstitium, and water movement out of the descending limb of Henle) is repeated until the full interstitial gradient is established (“countercurrent multiplier” concept). The osmolar concentration in the

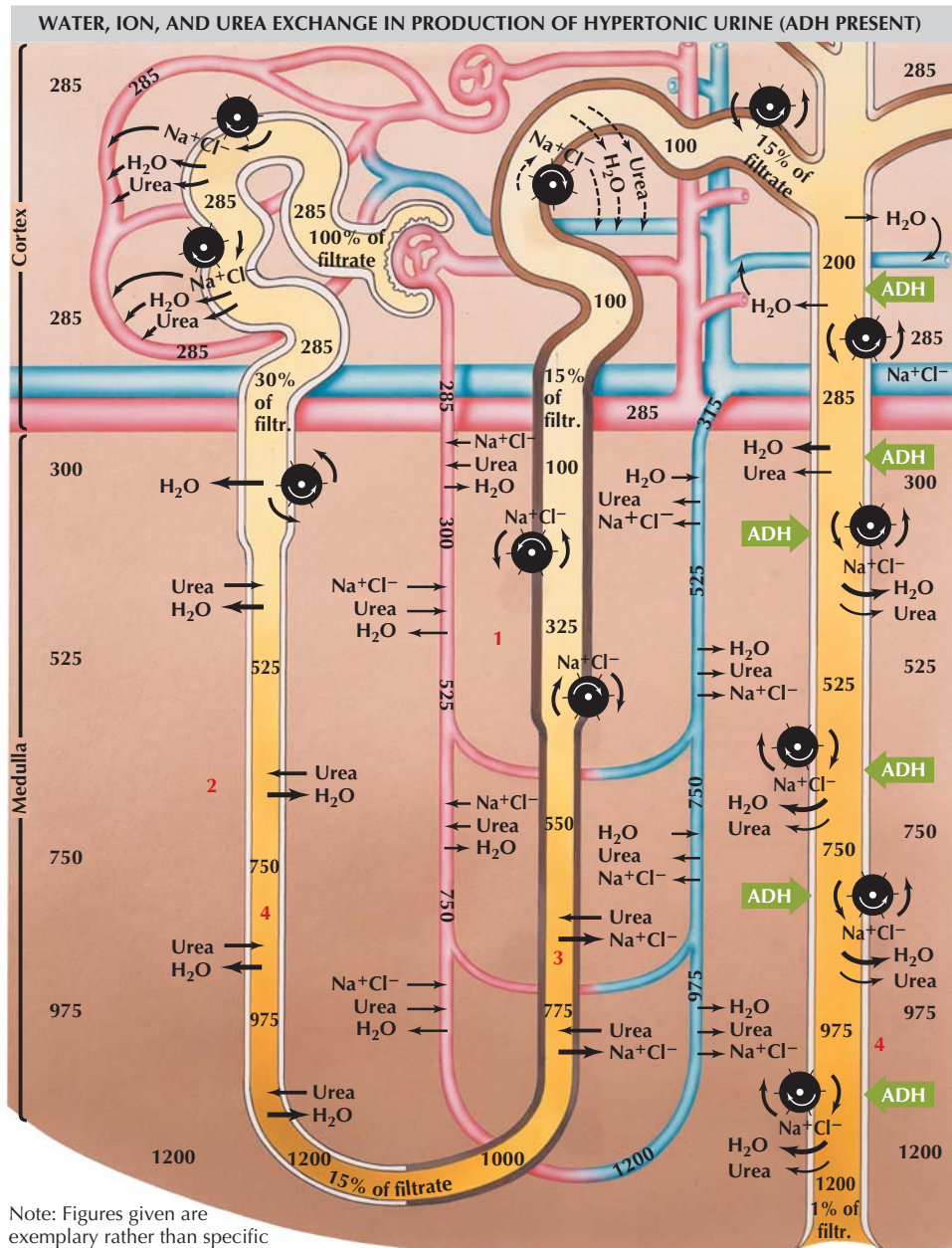


Figure 18.1 Medullary Interstitial and Tubular Concentration Gradients The concentration gradient is established by (1) transport of solutes, but not water, out of the thick ascending limb of Henle (TALH) via the $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ transporters (diluting limb); (2) free water reabsorption from the descending limb of Henle, which increases the tubular fluid osmolarity in the descending limb (concentrating limb); (3) as more tubular fluid flows from the descending limb to the TALH, the more concentrated tubular fluid allows further transport of solutes into the interstitium; and (4) urea recycling contributes to the gradient, because it remains in the tubular fluid in the loop of Henle contributing to the tubular fluid osmolarity while water is reabsorbed from the descending limb, and when ADH is present, both water and urea reabsorption increases in the medullary collecting ducts (CDs), and the urea is recycled to the inner medulla.

interstitium deep in the medulla is dependent on the length of the loops of Henle (the longer the loops, the higher the concentrating ability). In humans, the highest interstitial concentration (deep in the inner medulla) is ~ 1200 mosm/L. This allows the concentration of tubular fluid to reach ~ 1200 mosm/L at the bottom of the loops of Henle (see Fig. 18.1).

- Finally, *urea recycling* contributes to developing and maintaining the interstitial osmolar gradient, because
 - it remains in the tubular fluid while water is reabsorbed from the descending limb, contributing to the tubular fluid osmolarity, and
 - ADH increases both water *and* urea reabsorption in the medullary (but not cortical) collecting ducts (CD); the urea is recycled into the inner medulla, contributing to the interstitial concentration gradient.



The ability to concentrate urine differs between species. Animals that live in desert environments (desert rodents, camels) have exceptionally long loops of Henle and have the ability to concentrate their urine to more than 2000 mosm/L, allowing tremendous fluid retention. The medullary interstitium of these animals appears to have additional osmotic agents ($\sim 20\%$ extra “osmolytes” such as sorbitol and myo-inositol) to help accomplish this action.

Concentration of the Urine

As discussed in Chapter 1, to maintain plasma (and thus cellular) osmolarity, fluid volume must be controlled. Either a small increase in plasma osmolarity ($\sim 1\%$) or a significant decrease (a greater than 10% loss) in plasma volume (from, for example, hemorrhage or severe dehydration) will elicit release of antidiuretic hormone (ADH, also called vasopressin) from the posterior pituitary gland. This hormone binds to V2 receptors on principal cells of the renal collecting ducts (Fig. 18.2). The ADH increases apical water channels, or **aquaporins** (in the principal cells, AQP-2), which allow only water to be reabsorbed, effectively concentrating solutes in the tubular fluid, which at this point is considered urine.

The urine-concentrating mechanism depends on the plasma level of ADH and the osmolar concentration of the interstitial fluid surrounding the collecting ducts. Plasma ADH is tightly regulated, and water channels are continually being inserted and removed from the apical membranes of the CD cells, to maintain extracellular fluid balance through the regulated retention and excretion of water in the urine. The ability of the ADH to effectively promote water reabsorption is dependent on the medullary concentration gradient. ADH-dependent urea recycling plays an important role in the ability to concentrate urine, because the additional urea added to the

interstitium when ADH is elevated helps maintain the interstitial gradient for water reabsorption. Therefore, dehydration $\rightarrow$ higher plasma osmolarity $\rightarrow$ higher ADH levels $\rightarrow$ increased AQP and water reabsorption $\rightarrow$ more concentrated urine. Under extreme conditions, the urine can be concentrated maximally to 1200 mosm/L, with minimal urinary water loss.

DILUTION OF URINE

When there is excess extracellular fluid, plasma osmolarity is reduced and pituitary ADH release is inhibited. This affects water reabsorption in both the descending limb of Henle and the CDs. If less water is absorbed out of the tubular fluid in the descending limb, the tubular fluid cannot be concentrated to as high a level, and there is increased tubular fluid flow to the TALH, which reduces the amount of solutes that can be transported into the interstitium. This effectively disrupts the interstitial gradient, as illustrated in Figure 18.3. This decrease in ADH results in fewer water channels in the CDs, and less medullary water and urea reabsorption, producing diuresis (increased production of hypotonic urine). When the excess fluid is excreted, the plasma osmolarity will increase, stimulating ADH, and the interstitial concentration gradient will be reestablished over several hours.

FREE WATER CLEARANCE

Increased urine output in response to expansion of plasma volume entails both sodium excretion (natriuresis) and water excretion (diuresis). The ability to excrete hypotonic urine is as important to fluid homeostasis as the ability to excrete concentrated urine, and impairment of this mechanism can be life-threatening (see exercise-induced hyponatremia). The concept of free water clearance is useful in quantifying water excretion during diuresis; it is defined as water excretion in excess of the water required for iso-osmotic excretion of the solutes present in the urine. Free water clearance is



Micturition is the process of emptying the bladder (urination). Micturition is under voluntary control, because the external sphincter of the bladder is skeletal muscle. However, the micturition reflex system is under both sympathetic and parasympathetic control. While the bladder is filling, the sympathetic nerves relax the smooth muscle of the bladder wall, accommodating the urine, and contract the internal urethral sphincter smooth muscle. When the bladder becomes “full,” mechanoreceptors signal a spinal reflex arc that stimulates parasympathetic contraction of the bladder (detrusor muscle) and relaxation of internal sphincters. The external urethral sphincter is skeletal muscle, and is voluntarily relaxed, allowing urination.

MECHANISM OF ANTIDIURETIC HORMONE IN REGULATING URINE VOLUME AND CONCENTRATION

ADH is produced in supraoptic and paraventricular nuclei of hypothalamus and descends along nerve fibers to neurohypophysis, where it is stored for subsequent release.

Blood osmolality and volume are modified by fluid intake (oral or parenteral); water and electrolyte exchange with tissues, normal or pathological (edema); loss via gut (vomiting, diarrhea); loss into body cavities (ascites, effusion); or loss externally (hemorrhage, sweat).

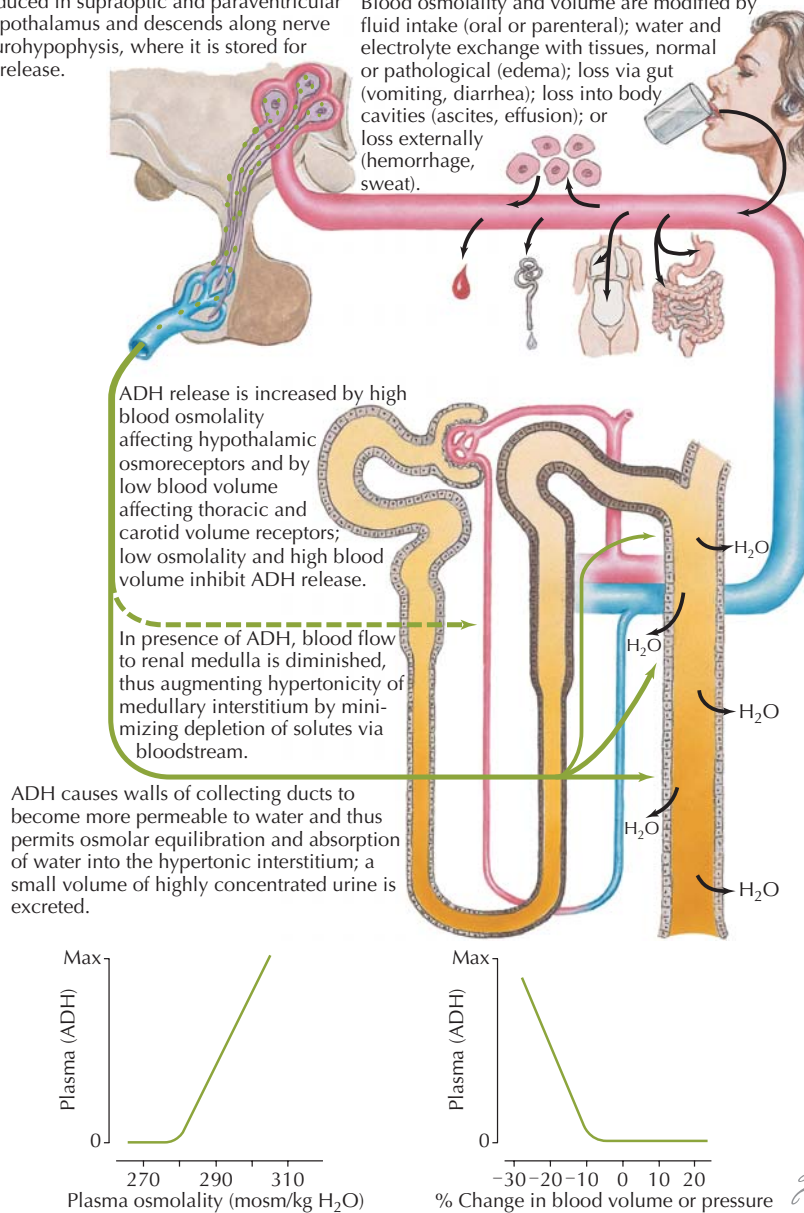


Figure 18.2 The Renal Response to ADH Secretion The scheme above illustrates the release and action of antidiuretic hormone (ADH). In response to dehydration, ADH is secreted from the posterior pituitary gland into the circulation. It acts on the kidneys to increase water channels in the collecting ducts, allowing solute-free water absorption, and also increases urea reabsorption in the medullary collecting ducts. The additional urea is added to the inner medullary interstitium and contributes to the high interstitial osmolar concentration.

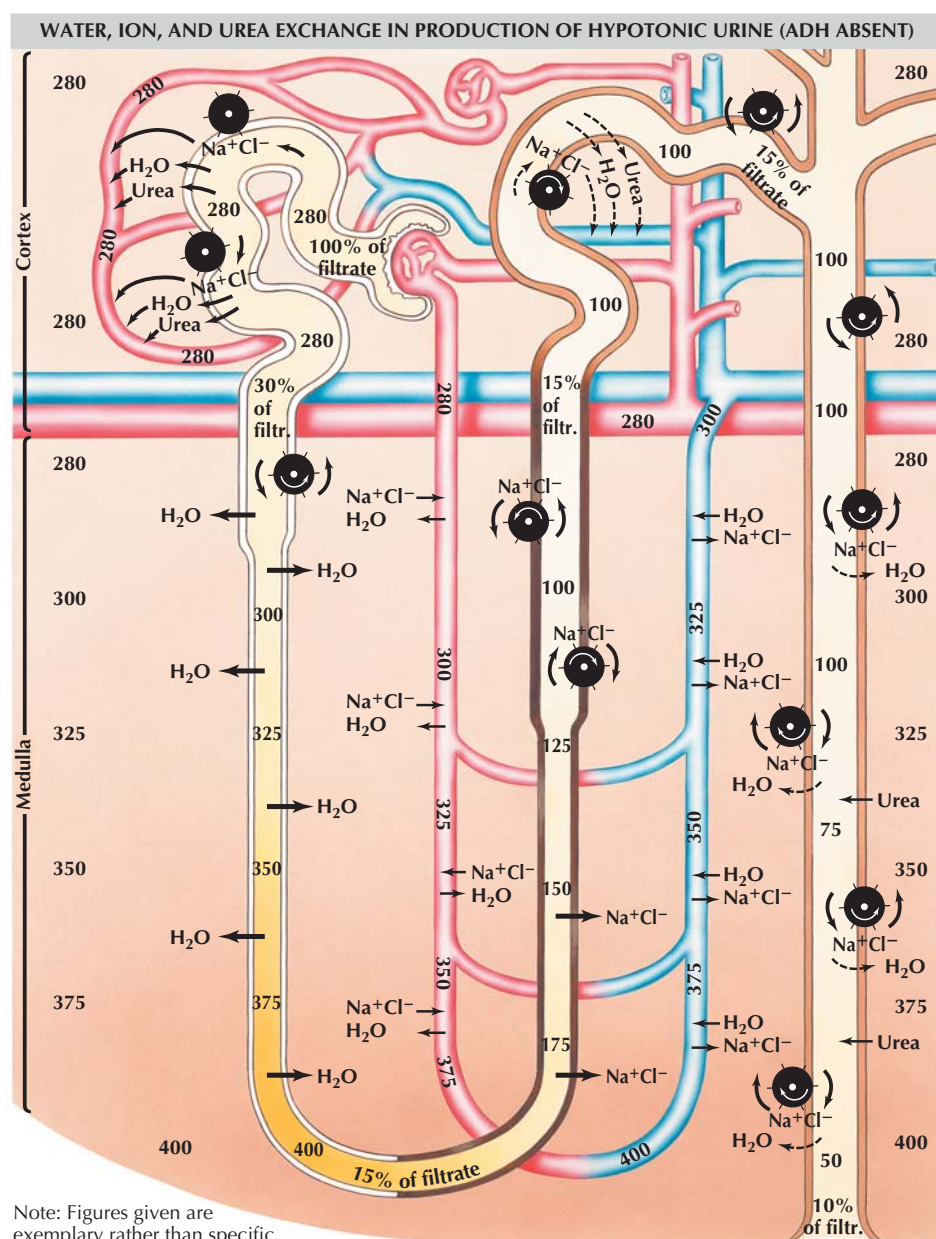


Figure 18.3 Dilution of Urine Excess extracellular fluid (ECF) will decrease antidiuretic hormone (ADH) secretion and reduce water channels in the collecting ducts. The disruption of the interstitial concentration gradient and lack of water channels produce diuresis.

determined by subtracting the osmolar clearance from the urine flow rate,

$$C_{H_2O} = \dot{V} - (U_{osm}/P_{osm}) \times \dot{V}$$

Thus, in dilute urine (e.g., U/P is less than 1) the C_{H_2O} is a positive number, implying that water was excreted. In

contrast, if the urine is concentrated (e.g., U/P is greater than 1), the C_{H_2O} is negative, implying that water was retained. If urine osmolarity equals plasma osmolarity, the C_{H_2O} equals zero.

CLINICAL CORRELATE

Chronic Pyelonephritis

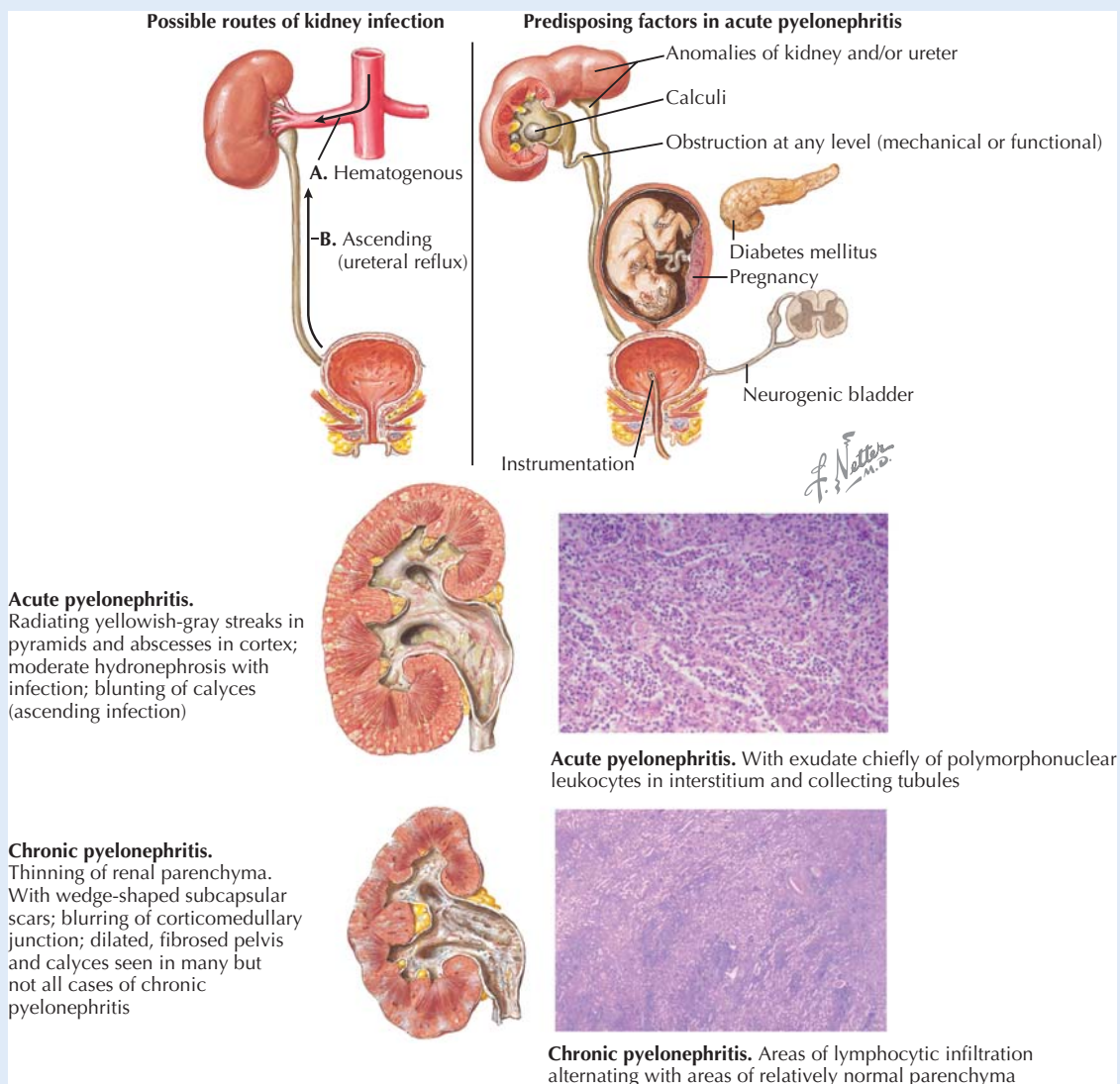
Pyelonephritis is inflammation of the renal pelvis, caused by bacterial infection. While acute kidney infections are usually caused by urinary tract infections (UTI), they can recur, and with each occurrence they can further damage the kidney. UTIs typically arise from contamination from bowel microorganisms, although with recurring infections that reach the kidney, potential underlying causes such as kidney stones or other anatomical abnormalities should be considered. Increased risk of pyelonephritis is associated with diabetes, pregnancy, prostate enlargement, compromised immunity, and sexual behavior and spermicide use.

The complications arising from **chronic pyelonephritis** relate to the area that is infected. The smooth muscle lining of the renal pelvis exhibits peristaltic activity that helps direct the newly col-

lected urine toward the ureters for transit to the urinary bladder. The infection causes abscesses and necrosis of the pelvic tissue, which can lead to fibrosis and scarring.

As more of the medulla (tubules and parenchyma) becomes damaged with repeated infections, the ability to maintain the interstitial concentration gradient is compromised. Because high interstitial osmolarity provides the gradient that allows ADH-dependent free water reabsorption and urine concentration, loss of the deep gradient restricts the ability to excrete concentrated urine and thus can cause polyuria, despite dietary water restriction. The loss of tubules also reduces glomerular filtration rate (GFR), and thus renal function as a whole is diminished.

Pyelonephritis is treated with antibiotics over a period of several weeks. Increased fluid intake is recommended to flush any lower urinary tract bacteria out though increased urine production.



Chronic and Acute Pyelonephritis

CHAPTER 19

Regulation of Extracellular Fluid Volume and Osmolarity

INTRARENAL REGULATION OF SODIUM AND FLUID REABSORPTION

Renal sodium handling is closely regulated as part of the important process of extracellular fluid (ECF) homeostasis. A number of intrarenal factors can alter sodium (and thus, fluid) reabsorption in response to changes in systemic volume status:

- **Glomerular filtration rate (GFR):** Increases in GFR will increase the filtered load of sodium, and because the percentage of sodium reabsorbed in the proximal tubule does not change, the absolute amount of sodium entering the loop of Henle increases. Assuming that fractional reabsorption of sodium in the distal segments is unchanged, more sodium will be excreted. Conversely, if GFR decreases, the absolute amount of sodium entering the loop of Henle will decrease, as will sodium excretion (if fractional reabsorption in later segments is unchanged).
- **Tubular fluid flow rate:** While the tubular fluid flow rate may change, compensatory mechanisms within the kidney serve to regulate it within a normal range. This is necessary, because low flow rates will result in decreased delivery of sodium to the loop of Henle, low osmolar gradient in the interstitium, and poor ability to reabsorb water in the collecting duct. On the other hand, rapid tubular flow rates will wash out the osmolar gradient in the medullary interstitium. For this reason, tubuloglomerular feedback mechanisms are important in controlling the flow rate. When tubular flow is rapid, the macula densa cells of the juxtaglomerular apparatus secrete a vasoactive substance (adenosine or ATP) into the interstitial fluid adjacent to the afferent arteriole. The afferent arteriole constricts, which reduces HP_{GC} , and thus decreases the pressure for filtration, reducing GFR and tubular flow.
- **Baroreceptors** located in the afferent arteriolar vessel walls respond to changes in blood pressure. When stretched, they stimulate juxtaglomerular cells, causing the release of renin (this is an intrarenal mechanism, and does not involve the CNS vasomotor center).
- **Medullary blood flow:** If blood flow in the vasa recta increases, the medullary interstitial concentration gradient will be reduced, resulting in decreased solute reabsorption ($Na^+ - K^+ - 2Cl^-$) in the thick ascending limb of

Henle (TALH) and decreased water reabsorption in the thin descending limb of Henle (tDLH). The reduction in sodium reabsorption in the loop of Henle will increase the delivery of sodium and fluid to the distal tubules and collecting ducts (CDs). Because the concentration gradient is reduced, urine will not be effectively concentrated, and natriuresis/diuresis will occur.

NEUROHUMORAL CONTROL OF RENAL SODIUM REABSORPTION

The kidneys constantly respond to changes in blood pressure and ECF volume status, and this regulation is accomplished by intrarenal controls noted earlier, as well as several important neural and humoral mechanisms:

- **Sympathetic nerves** increase sodium reabsorption through several mechanisms. Sympathetic nerves innervate afferent and efferent arterioles (via α -adrenergic receptors). During sympathetic stimulation, arterioles constrict, decreasing GFR, and thus decreasing sodium excretion. There is also direct sympathetic innervation of the proximal tubule and loop of Henle, which, when activated, stimulates sodium reabsorption.
- The **renin-angiotensin-aldosterone system (RAAS)** has an important role in regulation of the renal retention of sodium and water, and thus, in the regulation of ECF volume and solute composition. In response to low tubular sodium concentration and low tubular fluid flow, juxtaglomerular cells produce the proteolytic enzyme **renin** and secrete it into the afferent arterioles (Fig. 19.1). Renin cleaves angiotensinogen (a plasma protein secreted by the liver) to angiotensin (Ang) I, which is converted to **Ang II** by angiotensin-converting enzyme (ACE) in the lung (and other tissues); Ang II stimulates adrenal medullary release of the mineralocorticoid aldosterone. In the kidney, Ang II has dual effects: it directly stimulates sodium (and thus, water) reabsorption in the proximal tubule and has vasoconstrictor effects on afferent and efferent arterioles, which result in a lower GFR, and sodium retention.
- **Aldosterone** binds to the cytoplasmic mineralocorticoid receptors in the late distal tubules and collecting tubules and stimulates Na^+ and water retention (and K^+

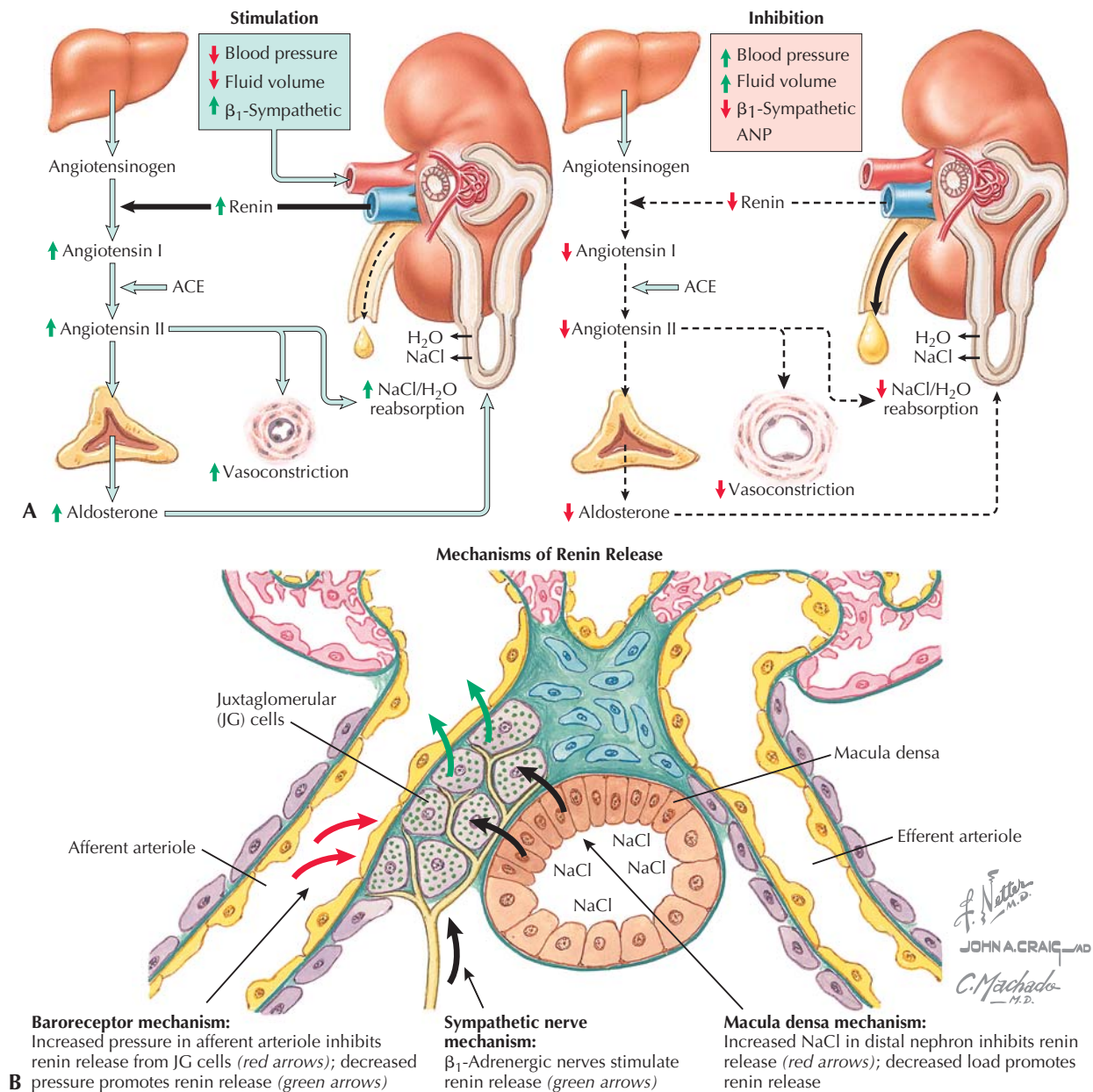


Figure 19.1 Mechanism of Renin Secretion and Factors Regulating the Renin-Angiotensin-Aldosterone System Renin is secreted from the juxtaglomerular cells in response to reduced sodium concentration and flow in the distal tubule (B). The cascade of events initiated to promote sodium and water reabsorption is illustrated in A.

secretion). Aldosterone increases apical Na⁺ channels and Na⁺/H⁺ antiporters and increases basolateral Na⁺/K⁺ ATPase activity. This further increases sodium and water retention and limits urinary losses.

- **Atrial natriuretic peptide (ANP)** is primarily produced by myocytes in the right atrium of the heart and is released into the blood in response to atrial stretch (as with increased blood volume). ANP opposes the actions

of Ang II by increasing GFR and inhibiting collecting tubule Na⁺ reabsorption. ANP causes natriuresis and diuresis, reducing ECF volume.

- **Urodilatin** is a natriuretic peptide related to ANP that is produced in the renal tubular cells and secreted *into the tubules* (not found in blood). The peptide acts at the collecting tubules to decrease Na⁺ reabsorption, resulting in natriuresis/diuresis.

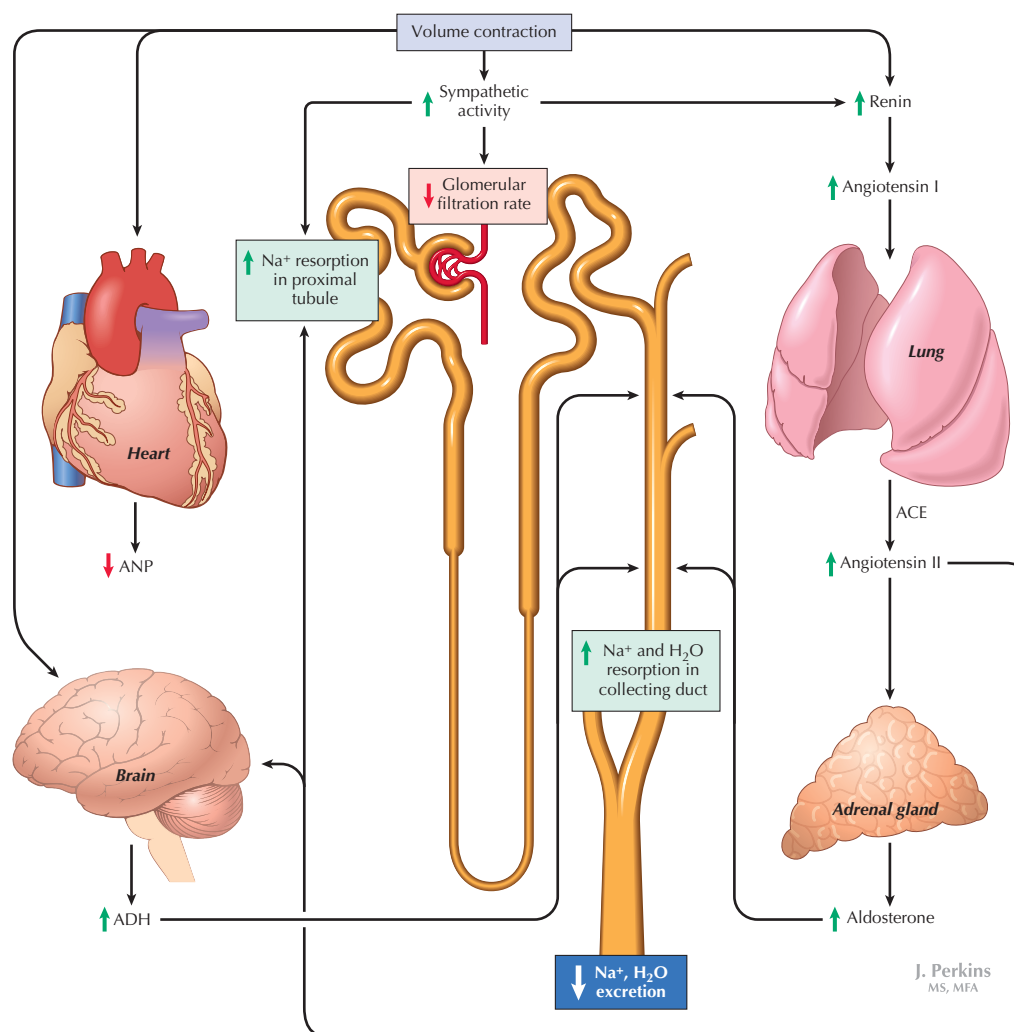


Figure 19.2 Renal Response to Volume Contraction In response to volume contraction (dehydration), the renin-angiotensin-aldosterone system is activated, stimulating renal sodium and fluid retention; antidiuretic hormone secretion from the anterior pituitary is stimulated to increase water reabsorption in the renal collecting ducts, and the sympathetic nerves are stimulated to increase renal sodium reabsorption and decrease glomerular filtration rate.

RENAL RESPONSE TO CHANGES IN PLASMA VOLUME AND OSMOLARITY

As discussed earlier, control of ECF is a continual process, with changes in plasma osmolarity and volume signaling multiple neural and hormonal systems to regulate the renal concentration and dilution of the urine. The integration of these systems is illustrated in the overall response to ECF volume contraction and expansion (Figs. 19.2 and 19.3). When plasma volume is contracted, fluid and sodium conservation systems are activated. During volume contraction, the kidneys respond to:

- Increased sympathetic nervous system (SNS) activity, which increases renal vascular resistance and decreases

GFR; proximal tubular sodium (and water) reabsorption increases.

- Activation of the RAAS, which increases Ang II and aldosterone, enhancing sodium (and water) reabsorption in the proximal tubules and CDs, respectively.
- The increase in antidiuretic hormone (ADH), which increases water channels in the CDs, enhancing solute-free water absorption.

These systems limit further volume contraction by decreasing the loss of fluid in urine. When plasma volume is expanded, these systems are reversed, allowing elimination of fluid and reduction of plasma volume and ECF (Fig. 19.3). Furthermore, the increase in ANP from the right

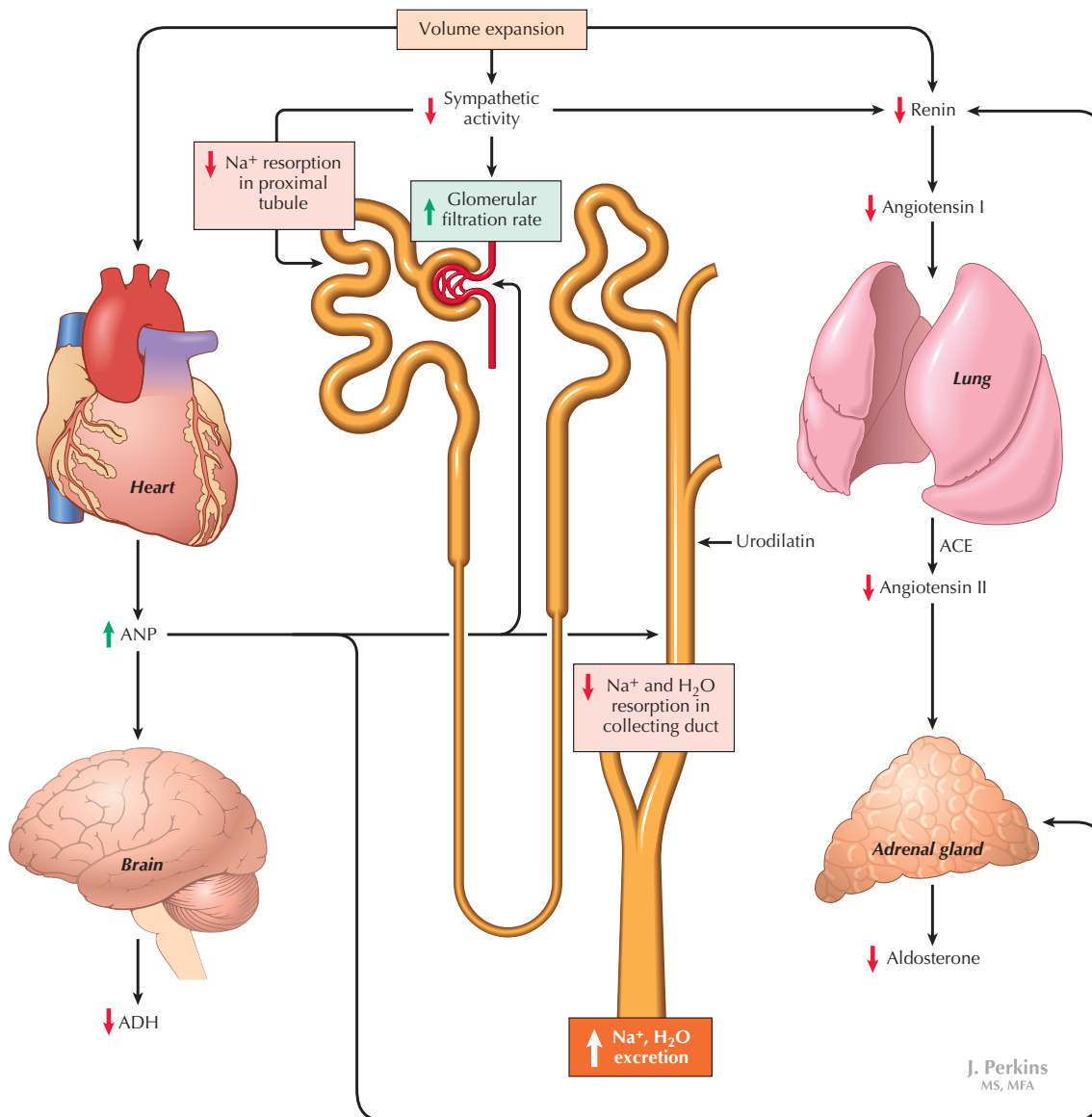


Figure 19.3 Renal Response to Volume Expansion In response to volume expansion, sodium- and fluid-retaining mechanisms are decreased (RAAS, ADH), and the increased stretch on the cardiac right atrium releases atrial natriuretic peptide, which acts at the kidneys to decrease sodium and water retention, creating a diuresis and natriuresis, eliminating the excess fluid.



Because of the potent vasoconstrictor and sodium-retaining effects of angiotensin II on the kidney, inhibition of Ang II is a major therapeutic intervention for treatment of hypertension. **Angiotensin-converting enzyme (ACE) inhibitors** (e.g., captopril, enalapril) prevent conversion of Ang I to Ang II, whereas **angiotensin receptor blockers (ARB)** (e.g., losartan, candesartan) block Ang II AT₁ receptors. Both interventions decrease systemic vascular resistance and increase urinary sodium and water excretion.



To maintain homeostasis, almost all of the filtered sodium must be reabsorbed—loss of even a few percent of the filtered sodium can result in severe sodium deficiency. Although the fine-tuning by aldosterone increases sodium reabsorption only about 2% to 3%, in aldosterone insufficiency (Addison's disease), these losses can lead to severe ECF sodium depletion, ECF volume contraction, and circulatory collapse.

cardiac atrium has a key role in producing natriuresis and diuresis by:

- Inhibiting aldosterone secretion.
- Decreasing ADH.

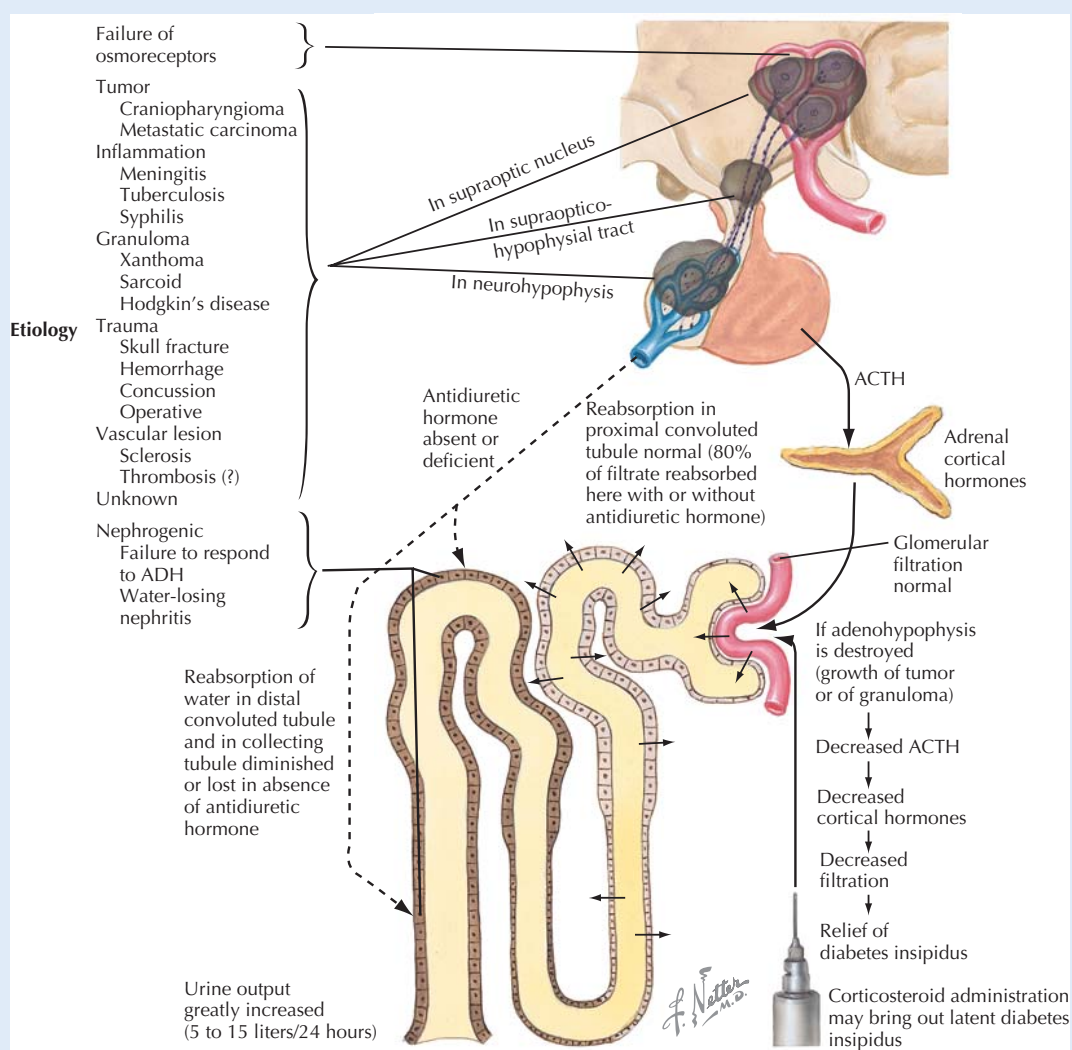
- Increasing GFR by relaxing mesangial cells (and also through renovascular effects).
- Decreasing sodium (and water) reabsorption in the CDs by reducing Na^+ channels (ENaC).

CLINICAL CORRELATE

Diabetes Insipidus

ADH allows the kidneys to concentrate urine. Insufficiency of ADH secretion results in **diabetes insipidus (DI)**, a disease in which large volumes of hypotonic urine are excreted. **Central diabetes insipidus** is usually caused by trauma, disease, or surgery affecting the posterior pituitary gland or hypothalamus. **Nephrogenic DI** is rare and involves a reduction in ADH V2 receptors or reduction in the AQP2 water channels in the collecting ducts of the kidney, reducing sensitivity of the cells to ADH.

Fluid intake must increase to compensate for the urinary losses, which can range from 3 to 18 liters per day. Mortality is rare, although children and elderly people are at greater risk, from severe dehydration, cardiovascular collapse, and hypernatremia. ADH analogs such as DDAVP (e.g., desmopressin) are used to treat central DI. The analogs act like endogenous ADH and increase water channels in the collecting ducts. Nephrogenic DI can respond to indomethacin as well as dihydrochlorothiazide, which is a diuretic that has a paradoxical effect to increase water reabsorption in DI.



Central Diabetes Insipidus The causes of central diabetes insipidus, with the effects on the kidneys, are depicted in this scheme.

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CHAPTER 20

Regulation of Acid–Base Balance by the Kidneys

CONTROL OF EXTRACELLULAR FLUID pH

Why is systemic acid–base status important, and what is the role of the kidneys? As noted in Chapter 15, the blood (and extracellular fluid [ECF]) pH must be maintained within a narrow range (7.35 to 7.45) to allow normal cellular functions. This physiologic pH range corresponds to a narrow range in H^+ concentration (45 to 35 nanomoles per liter [nM/L]). The narrow physiologic window for H^+ concentration demonstrates the necessity for tight control of pH at all times.

Because acid balance is critical for controlling ECF pH, the daily entry of H^+ into the ECF must equal the losses. Net gain of H^+ can occur from ingestion of acid contained in foods (acidic drinks, proteins), cell and protein metabolism, hypoventilation, and diarrhea (loss of HCO_3^- results in gain in H^+ , described later). Net losses of acid can result from hyperventilation, vomiting, and of course urinary acid excretion. Under normal conditions, the daily gain of acid from normal metabolism (sulfuric acid, phosphoric acid, keto acids, etc.) and diet (proteins) will be equaled by acid excreted in the urine. CO_2 is a volatile acid and can be excreted by the lungs, but when it is in the blood it is dissolved, and contributes to the overall acid pool. A general scheme is illustrated in Figure 20.1.

In general, we ingest and produce ~40 to 80 *millimoles* (mM) of acid each day. This is a huge amount compared with the ~40 *nanomolar* level (at pH 7.4) that is maintained in the ECF. This excess acid must be:

- **Buffered**, to prevent a fall in pH to below physiologic levels (e.g., below 7.35).
- **Excreted** in the urine, which is the job of the kidneys.

Buffering of Acid

The body handles excess acid using intracellular and extracellular buffers, in a continuous regulatory “dance,” escorting acid in and out of cells and through the blood to the kidneys for excretion.

Extracellular Buffering

Bicarbonate (HCO_3^-) is the major ECF buffer, available for consuming free H^+ through the following reaction:



The carbonic acid can be converted to CO_2 and H_2O , in the presence of carbonic anhydrase (CA). This occurs in ECF and tissues, allowing diffusion of CO_2 and H_2O into and out of tissues. CO_2 does not normally contribute to the *net gain* in acid (because the H^+ gained in the above reaction is utilized, as water is formed in the lungs when the CO_2 is blown off during respiration). However, CO_2 does contribute to the net acid gain during **hypoventilation**.

As described in Chapter 15, the **Henderson-Hasselbalch** equation describes the relation between acid–base status and pH as:

$$pH = 6.1 + \log[\text{base/acid}]$$

where the base is plasma bicarbonate (normally ~24 mM/L of ECF), and acid is the $PCO_2 \times 0.03$ (solubility constant) (normally 40 mm Hg $\times 0.03 = 1.2$ mM/L of ECF). And thus, under normal conditions,

$$pH = 6.1 + \log[24/1.2] \\ pH = 7.4$$

The kidneys control the amount of base (free bicarbonate) in the ECF. This is accomplished by generating new bicarbonate, or excreting excess bicarbonate (in alkalosis). Although the amount of acid is also controlled by urinary acid excretion, in acidosis and alkalosis respiration can also regulate ECF acid. ECF phosphates and proteins also contribute to the buffering, but to a very small extent.



What makes a good buffer? Good buffers have a pK that is close to the physiologic pH of 7.4. In the ECF, the best buffer would be phosphate (HPO_4^{2-}), which has a pK of 6.8 (which at a pH of 7.4 results in a base to acid ratio of 4:1 [HPO_4^{2-} : $H_2PO_4^-$]). However, there is relatively little phosphate in the ECF (~1 mM/L), so it is not an effective ECF buffer. Instead, bicarbonate (pK = 6.1 and base to acid ratio is 20:1 [HCO_3^- : H_2CO_3] at pH of 7.4) is the main ECF buffer, because of its high ECF concentration (~24 mEq/L). The large amount of free bicarbonate allows ready buffering of additional acid load.

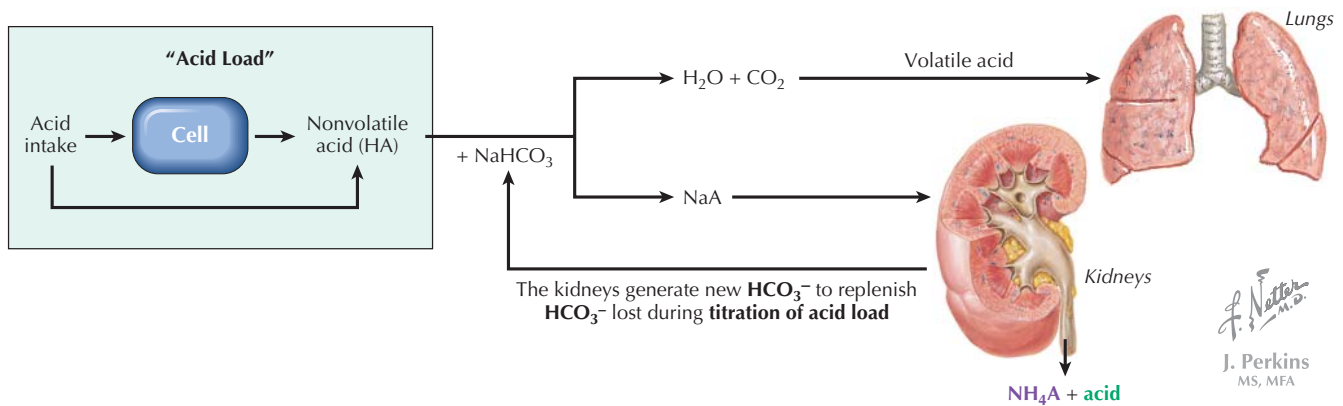


Figure 20.1 General Scheme for Eliminating Excess Acid New acid is added daily from diet and metabolism and must be buffered (primarily by ECF bicarbonate), and then excreted in the urine.

Intracellular Buffering

Although phosphates offer minor buffering in the ECF, when needed they provide major buffering capacity *within* the cells, because of their high concentration. In addition, cellular proteins contribute to the buffering process. Movement of H^+ into and out of cells occurs through cation exchange (H^+/K^+ and Na^+/H^+ antiporters). The buffering process minimizes the effects of generated and ingested acid on pH of the ECF and allows shuttling of acid to the kidneys, where it can be excreted.

HCO_3^- AND H^+ HANDLING THROUGH THE RENAL TUBULE

HCO_3^-

As previously described, the process of bicarbonate reabsorption occurs in the proximal tubule, thick ascending limb of Henle (TALH), and collecting duct and is dependent on the secretion of H^+ into the tubular lumen (see Fig. 17.3). This cycle effectively reclaims 100% of the filtered bicarbonate back into the ECF, and under normal conditions there is no urinary excretion of bicarbonate.

The exception to this occurs in alkalosis, when acid–base balance depends on the removal of HCO_3^- . This is accomplished in the collecting ducts (CD), where the **β -intercalated cells** have the ability to secrete HCO_3^- into the tubule for excretion, via HCO_3^-/Cl^- antiporters. Again, this transporter is only active during alkalosis, when its activation results in HCO_3^- loss and H^+ accumulation.

H^+

H^+ is secreted into distal segments of the nephron in excess of filtered bicarbonate, with significant secretion occurring from the **α -intercalated cells** of the collecting ducts (see Fig. 17.3,

upper right panel). These cells have luminal H^+ ATPase pumps and actively secrete H^+ into the tubular fluid. One of the pumps is the H^+/K^+ pump, discussed in Chapter 17, which contributes to net potassium reabsorption during potassium depletion. The H^+/K^+ ATPase is aldosterone-sensitive. The basolateral membranes have HCO_3^-/Cl^- exchangers that transport HCO_3^- into the interstitium, from which it enters the blood. Excess acid secreted along the nephron must be buffered (to allow continued secretion of H^+) and excreted. Factors that can regulate H^+ secretion in the nephron are given in Table 20.1.

RENAL MECHANISMS CONTRIBUTING TO NET ACID EXCRETION

The ingested/generated acid load is always buffered in the plasma and is then dissociated from the buffers in the kidneys in the process of bicarbonate metabolism. The free H^+ is secreted into the tubular lumen (in PT, TALH, CD), where it is incorporated into either phosphates (to become a **titratable acid**), or **ammonia** (to become **ammonium**). This buffers the acid controlling urine pH.

A key concept is that a new bicarbonate ion is generated for every H^+ ion that is excreted, and this is always in a 1:1 ratio (1 HCO_3^- reabsorbed to 1 H^+ excreted). This occurs because both the secretion of H^+ into the renal tubule and excretion of ammonium (NH_4^+) result in HCO_3^- reabsorption (Fig. 20.2). If the secreted H^+ ends up being excreted as a titratable acid or ammonium, the amount excreted is directly equated with generation of new bicarbonate.

Production of Titratable Acids

The primary form of titratable acid is phosphoric acid ($H_2PO_4^-$). Remember, phosphate is a strong buffer (pK 6.8) but is not readily available in the ECF, because of its low

Table 20.1 Factors Influencing H⁺ Secretion by the Nephron

Factor	Principal Site of Action
INCREASED H⁺ SECRETION—PRIMARY	
↓HCO ₃ [−] concentration (↓pH)	Entire nephron
↑Arterial PCO ₂	Entire nephron
INCREASED H⁺ SECRETION—SECONDARY	
↑Filtered load of HCO ₃ [−]	Proximal tubule
↓ECF volume	Proximal tubule
↑Angiotensin II	Proximal tubule
↑Aldosterone	Collecting duct
Hypokalemia	Proximal tubule
DECREASED H⁺ SECRETION—PRIMARY	
↑HCO ₃ [−] concentration (↑pH)	Entire nephron
↓Arterial PCO ₂	Entire nephron
DECREASED H⁺ SECRETION—SECONDARY	
↓Filtered load of HCO ₃ [−]	Proximal tubule
↑ECF volume	Proximal tubule
↓Aldosterone	Collecting duct
Hyperkalemia	Proximal tubule

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concentration (~1 mM/L). However, in the tubular fluid, the amount of filtered phosphate (FLPi) is significant (FLPi = 1 mM/L × ~140 L/day (GFR) = ~140 mM Pi/day), and part of this can be used for buffering and excreting H⁺ (bicarbonate cannot be used, because it is completely reabsorbed). The amount of phosphate available to form titratable acid depends on (1) the amount of basic phosphate (HPO₄^{2−}) available to bind with H⁺, and (2) the renal handling of phosphate.

- **Phosphate buffering:** According to the Henderson-Hasselbalch equation, at a blood pH of 7.4, there will be 4 times the amount of base (HPO₄^{2−}) to acid (H₂PO₄[−]), and this base is available for buffering excess H⁺. Thus, the ~140 mM/day of filtered phosphate includes about 116 mM/day of HPO₄^{2−} that could be used as buffer—however, not all of this is available because of the tubular handling of phosphate.

- **Renal handling of phosphate:** In the normal adult, ~75% of the filtered phosphate is reabsorbed and therefore unavailable for generating titratable acid. Thus only 25% of the filtered HPO₄^{2−} can be used, or ~29 mM/day (116 mM/day × 0.25 = 29 mM/day).

Formation of titratable acids occurs in the CDs at the α-intercalated cells. At this point of the nephron, phosphate reabsorption is complete and the final reabsorption of HCO₃[−] is occurring. Thus, the excess H⁺ that is actively secreted can bind to the HPO₄^{2−}, creating the titratable acid H₂PO₄[−], which is then excreted. Under normal conditions, titratable acids will be the main source of acid excretion, but their maximal rate of excretion is fixed because titratable acid formation depends on the amount of phosphate reabsorption occurring, and this is not enough to eliminate the daily acid load. The remaining acid is buffered by NH₃, and when the acid load increases, ammoniogenesis will be further stimulated to handle the load.

Ammoniogenesis

The proximal tubule cells are capable of producing ammonia from glutamine, extracted from the tubular fluid as well as peritubular capillary blood. The ammonia can buffer H⁺ by forming ammonium (NH₄⁺) that can ultimately be excreted in the urine. A key aspect of this reaction is that the excretion of NH₄⁺ produces new HCO₃[−] that is reabsorbed into the plasma.

In the proximal tubular cells, the glutamine is hydrolyzed to produce glutamate and one NH₃. The glutamate is further metabolized to α-ketoglutarate, producing another NH₃. The two NH₃ are immediately combined with two H⁺, forming two NH₄⁺. Additional α-ketoglutarate metabolism yields two HCO₃[−]. Thus, a single glutamine generates two HCO₃[−], which are reabsorbed as *new* HCO₃[−], and two NH₄⁺, which are secreted into the tubular fluid (see Fig. 20.2, *upper right*).

The NH₄⁺ produced in the proximal tubule is not directly excreted. Instead it is reabsorbed in place of K⁺ by the Na⁺-K⁺-2Cl[−] transporters. The NH₃ stays in the interstitial fluid, increasing the medullary interstitial concentration of NH₃. The dissociated H⁺ is secreted into Henle's loop in exchange for Na⁺. The NH₃ gradient promotes secretion of NH₃ into the tubular lumen of the collecting ducts; NH₃ immediately binds free H⁺ in the collecting ducts, re-forming NH₄⁺ that is excreted in the urine (see Fig. 20.2, *lower right*).

Net Acid Excretion

Acid balance is determined by the difference between acid intake and urinary excretion of acid. Under normal conditions, intake will equal excretion. **Net acid excretion (NAE)** describes the total amount of acid that is excreted in the urine:

$$\text{NAE} = \text{TA} + \text{NH}_4^+ - \text{HCO}_3^-$$

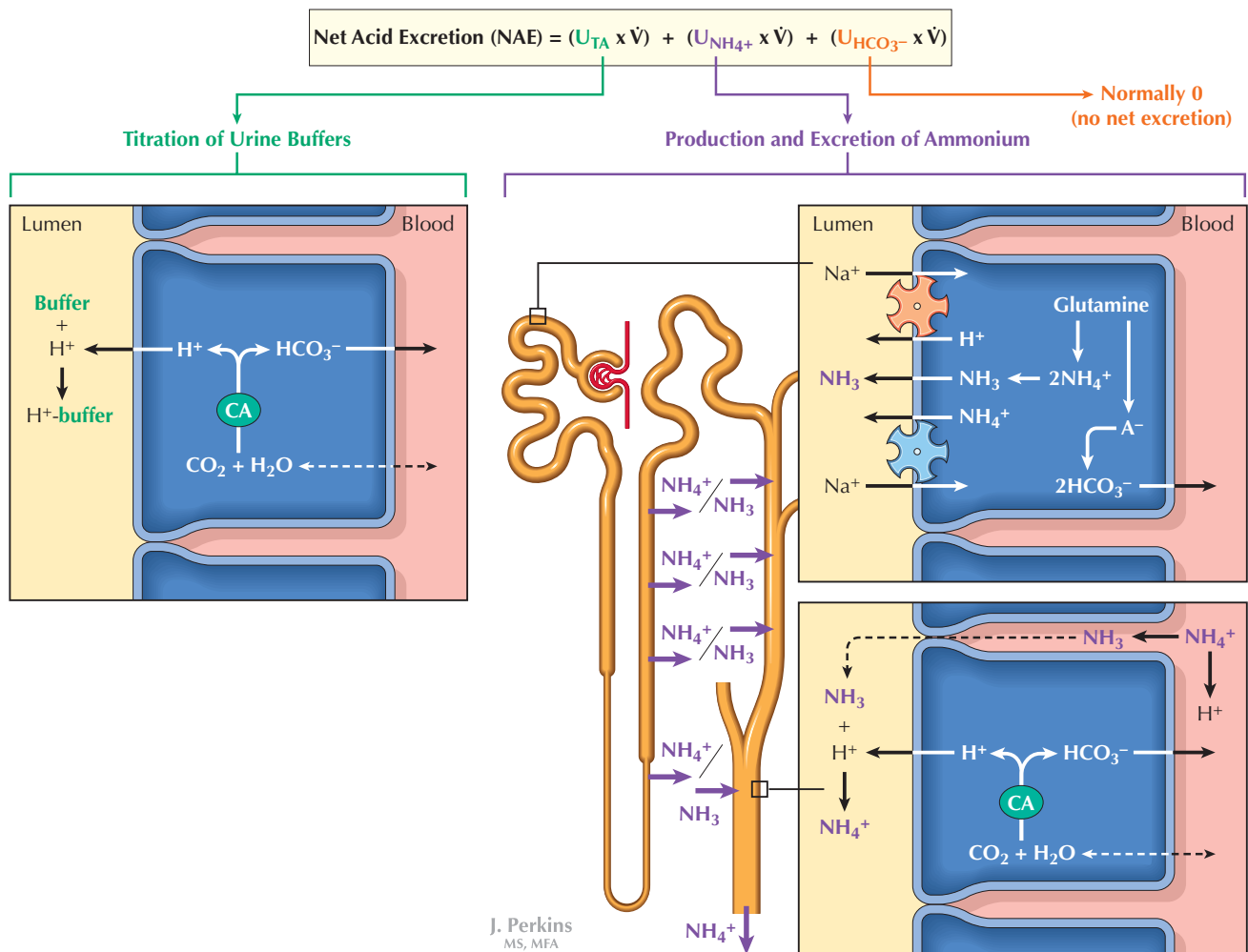


Figure 20.2 Renal Handling of Acid Excretion H⁺ is excreted as titratable acids (mainly phosphoric acids) and ammonium (NH₄⁺). With either mechanism, excretion of an H⁺ results in generation of a new HCO₃⁻ that enters into the blood.

where TA is titratable acids. Under most conditions, urinary HCO₃⁻ is zero (all HCO₃⁻ is normally reabsorbed). However, when HCO₃⁻ appears in the urine, it implies that H⁺ was added to the ECF (recall the 1:1 relationship between bicarbonate reabsorbed or excreted and H⁺ transported in the opposite direction). In the equation above, the HCO₃⁻ is subtracted from the TA and NH₄⁺ to account for the newly accumulated acid. HCO₃⁻ excretion is indicative of alkalosis or renal tubular acidosis, and in both conditions, there is an equimolar gain of acid for the HCO₃⁻ excreted.

Urine pH

Normal urine pH varies between 4.4 and 8 according to the acid-base status, with average values around 5.5 to 6.5. The minimal attainable urine pH is 4.4, which represents

1000-fold greater concentration of H⁺ than the blood pH of 7.4. This is the greatest concentration difference against which the H⁺ pumps can effectively secrete H⁺. This maximal level of urine acidity is only attained with severe metabolic acidosis.

ACIDOSIS AND ALKALOSIS

When pH falls outside of the normal physiologic range, acidosis (pH less than 7.35) or alkalosis (pH greater than 7.45) results. A disturbance is designated as:

- **Respiratory** (acidosis or alkalosis) if it is caused by abnormal CO₂, or
- **Metabolic** (acidosis or alkalosis), if the pH change is consistent with the alteration in HCO₃⁻.

Table 20.2 Sample Values for Uncompensated Acid–Base Imbalances

	pH	PCO ₂ (mm Hg)	HCO ₃ [−] (mM or mEq/L)
Primary respiratory acidosis	7.32	50	24
Primary metabolic acidosis	7.32	40	18
Primary respiratory alkalosis	7.54	28	24
Primary metabolic alkalosis	7.54	40	34

The acid–base status is assessed by examining the plasma values of pH, PCO₂, and HCO₃[−] (the *arterial blood gases*), which are the key components of the Henderson-Hasselbalch equation. Under normal conditions in arterial blood, these values will be:

	pH	PCO ₂ (mm Hg)	HCO ₃ [−] (mM)
Normal	7.4	40	24

When pH is altered, the primary disturbance can be identified by determining which component (PCO₂ or HCO₃[−]) is altered in the direction consistent with the change in pH. Increased PCO₂ or decreased HCO₃[−] will produce respiratory and metabolic acidosis, respectively; decreased PCO₂ or increased HCO₃[−] will produce respiratory and metabolic alkalosis, respectively (Table 20.2 and Fig. 20.3).

In general, compensation for metabolic disturbances will include altered respiratory rate: hyperventilation to blow off excess CO₂ in acidosis, or hypoventilation to retain CO₂ in alkalosis (discussed in Chapter 15). Compensation for respiratory disturbances will include altered renal bicarbonate and acid handling. **Metabolic compensation** occurs over several hours through the renal excretion of acid (in acidosis), or bicarbonate (in alkalosis), to return the pH toward normal. The examples in Table 20.2 represent *uncompensated* disturbances. The following paragraphs will focus mainly on metabolic disturbances.

Acidosis

Acidosis can result from either gain of acid or loss of bicarbonate. Net acid gain can arise from either decreased respiration (increasing CO₂), or from the accumulation of acids from metabolic sources (**metabolic acidosis**) outlined in the following text:

- Keto acids, from β -oxidation of fatty acids, which occurs during starvation and poorly controlled diabetes.
- Phosphoric acid, in renal failure, as a result of the inability to excrete the acids due to low GFR.
- Lactic acid, which is released from damaged tissues during hypoxia or heart failure.
- Ingestion of substances such as antifreeze (ethylene glycol) and Sterno® (containing methanol).

Bicarbonate losses are always caused by metabolic disturbances and mainly occur as a result of:

- Increased fecal elimination with prolonged diarrhea, because HCO₃[−] is secreted as a buffer and is lost with other constituents of the chyme (including the HCO₃[−] that is also secreted into the ileum and colon).
- Increased urinary excretion because of insufficient HCO₃[−] reabsorption in the proximal tubule.
- Elevated H⁺ secretion into the collecting ducts—this causes HCO₃[−] secretion from the β -intercalated cells of the collecting ducts to buffer the urine (distal renal tubular acidosis).

Whether due to addition of acid or loss of bicarbonate, the resulting acid load must be buffered systemically and then excreted by the kidneys. To compensate for acid gain, the plasma level of free bicarbonate falls as it forms H₂CO₃ to buffer the free acid. When pH is under 7.35, respiration will increase to “blow off” CO₂ and reduce acid load. Chronically, the main compensation will occur in the kidneys—increased acid secreted into the renal tubules will combine with phosphates and ammonia to form titratable acids and ammonium for excretion (see Fig. 20.2).

Although there can be an immediate increase in titratable acids because some excess HPO₄^{2−} is present, the ability to excrete titratable acids is limited by the amount of phosphate reabsorbed. The main urinary buffering of H⁺ is by ammonia (forming NH₄⁺). Ammoniogenesis increases during acidosis over a period of hours to days, allowing a great increase in ammonium excretion. The increase in NAE will increase pH toward normal values, although in severe cases of diabetic ketoacidosis, the kidneys may not be able to attain normal pH values until blood glucose is restored.

ANION GAP

The anion gap is used to differentiate between acidosis resulting from acid gain and acidosis caused by bicarbonate loss. This diagnostic tool utilizes the fact that the only way to lose bicarbonate is through HCO₃[−]/Cl[−] exchangers located in the β -intercalated cells of the collecting ducts and in the lower GI tract. Thus, if HCO₃[−] is lost, Cl[−] is gained, and this is reflected in the plasma concentrations of these anions. The anion gap (AG) is the difference in concentration between the

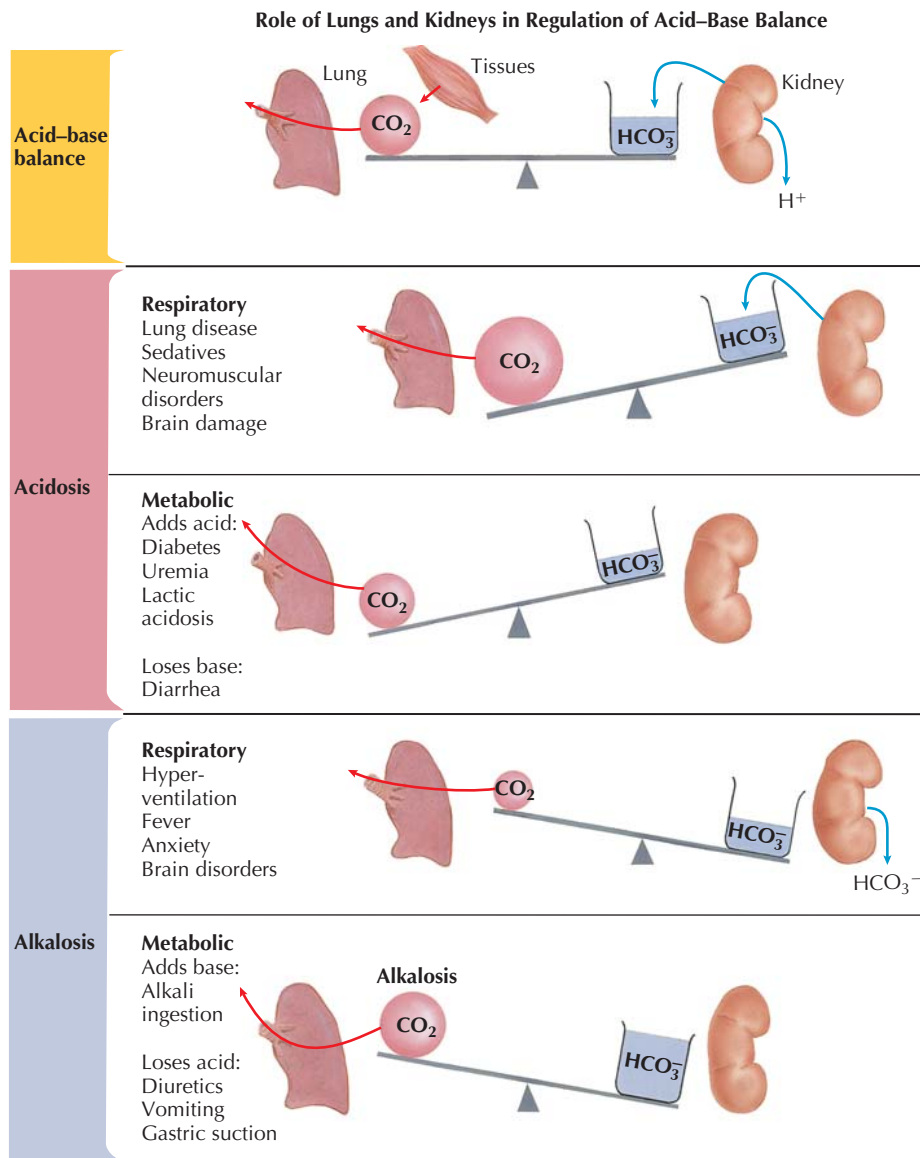


Figure 20.3 Role of the Lungs and Kidneys in the Regulation of Acid-Base Balance This scheme illustrates the changes in acid and base status in primary respiratory and metabolic disturbances. Under normal conditions (*upper panel*), the acid-base status is in balance, with kidneys absorbing all filtered HCO_3^- , and the daily gain of acid excreted by the kidneys. In both respiratory and metabolic imbalances, acid gain (acidosis) or loss (alkalosis) is compensated for by the renal excretion of acid or HCO_3^- , respectively. When blood pH is outside of the physiologic window (less than 7.35 or above 7.45), there is also respiratory compensation by hyperventilation or hypoventilation.

major cation, Na^+ , and the major plasma anions, Cl^- and HCO_3^- . When Cl^- and HCO_3^- concentrations are subtracted from the Na^+ concentration, the “anion gap” is normally ~8 to 12 mEq/L. The anion gap represents the sum of the concentrations of about 10 anions present in the plasma and includes proteins, lactate, citrate, phosphates, sulfates, etc.:

$$\text{AG} = \text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$$

As illustrated in Figure 20.4, *middle panel*, when acidosis results from acid loading, the anion gap increases, as a result of the lower plasma bicarbonate (used to buffer the acid). In contrast, when HCO_3^- is lost, it is replaced by Cl^- , and the anion gap is normal (see Fig. 20.4, *right panel*). Thus, the anion gap can help determine the cause of the metabolic acidosis, with the specific cause of the acid load or base loss remaining to be determined.

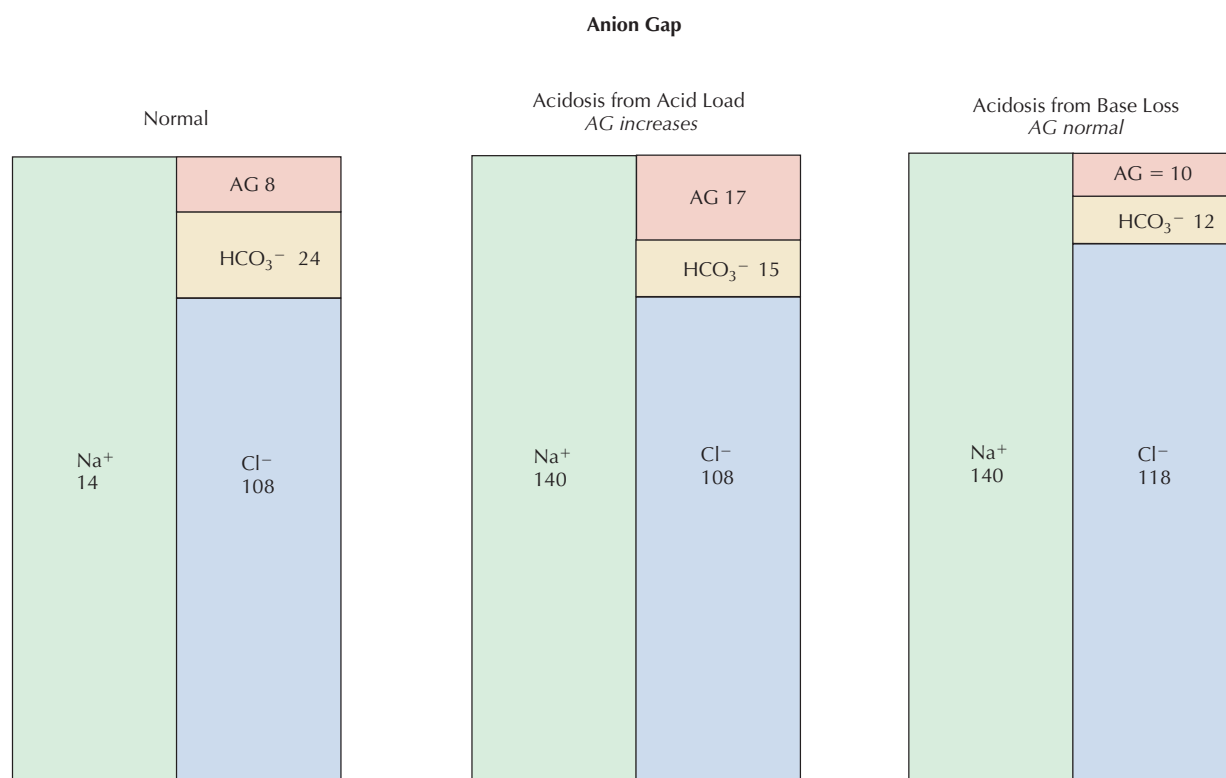


Figure 20.4 Plasma Anion Gap In cases of metabolic acidosis, measurement of the anion gap will reveal whether the disruption in pH was from acid loading or base loss.



Why don't the renal $\text{HCO}_3^-/\text{Cl}^-$ exchangers always work during alkalosis? It is important to recall that under normal conditions the $\text{HCO}_3^-/\text{Cl}^-$ exchangers in the β -intercalated cells are not active, because HCO_3^- is being retained, not excreted. However, these exchangers are expressed during alkalosis and depend on the luminal concentration of Cl^- to effectively transport the HCO_3^- into the tubule. The importance of distal luminal Cl^- is illustrated in contraction alkalosis, as seen with chronic vomiting and volume depletion.

In response to the alkalosis caused by loss of H^+ during vomiting, the renal $\text{HCO}_3^-/\text{Cl}^-$ exchangers are expressed but are unable to function because of the low level of Cl^- in the collecting duct fluid. The low Cl^- is a result of fluid- and sodium-retaining mechanisms (e.g., sympathetic nervous system [SNS], renin-angiotensin system, aldosterone, antidiuretic hormone [ADH]) stimulated in response to volume depletion. Chloride is avidly reabsorbed with the sodium and water, leaving little in the distal areas of the nephron. The problem can be corrected by isotonic fluid replacement, which will decrease SNS and renin-angiotensin system activation and aldosterone and ADH release, and increase Na^+ and Cl^- delivery to the collecting ducts. The $\text{HCO}_3^-/\text{Cl}^-$ exchanger will then be able to function, and excess HCO_3^- will be excreted, correcting the alkalosis.

Alkalosis

Alkalosis results from the loss of fixed acid or gain of HCO_3^- . Chronic hyperventilation (for example, at high altitude) will cause the loss of CO_2 and result in respiratory alkalosis. **Metabolic alkalosis** can result from:

- Vomiting, which results in the loss of significant amounts of H^+ in the form of HCl and volume contraction.
- Sodium bicarbonate overdose.
- Chronic diuretic usage, which can cause hypokalemia, volume contraction, aldosterone excess, and chloride depletion, all of which are associated with alkalosis.

During alkalosis, $\text{HCO}_3^-/\text{Cl}^-$ exchange by transporters in β -intercalated cells of the renal collecting ducts is activated, allowing secretion and excretion of bicarbonate until normal pH is attained.

CLINICAL CORRELATE
Metabolic Ketoacidosis

A common cause of ketoacidosis is poorly controlled diabetes. When glucose is not able to efficiently enter cells there are two immediate consequences: elevated blood glucose, and increased cellular β -oxidation of fatty acids to provide an alternate energy source in the absence of intracellular glucose. If the rate of oxidation is high, metabolites including ketoacids (or ketone bodies) enter the blood, decreasing the pH. Several problems ensue:

- **Acidosis:** The ketoacids are filtered at the glomeruli and eliminated in the urine, but the production overwhelms excretion. Plasma bicarbonate may not be sufficient to buffer the acid load, and in severe cases the free bicarbonate will be heavily consumed, and blood pH will be dangerously reduced (i.e., below 7.2). Contributing to this problem, ketone bodies interfere with the production of ammonia from glutamine in the nephron, and thus decrease new bicarbonate generation (further decreasing plasma bicarbonate levels).
- **Glucosuria:** Glucose is excreted in the urine, because the elevation in filtered glucose load overwhelms the renal sodium-glucose transporters. The glucose pulls water with it, creating polyuria (high urine volume).
- **Volume expansion and cellular contraction:** When glucose increases in the blood, it exerts additional osmotic force, keeping water in the vascular space. This expands the ECF, while the ICF decreases. Blood pressure may increase, and the concentration of other plasma constituents, including electrolytes and proteins, will be diluted. Ultimately, diabetic ketoacidosis may actually result in dehydration due to osmotic diuresis and vomiting.
- **Toxicity:** The keto acids in the blood are toxic and ultimately may cause coma and death if blood glucose level is not corrected. The most severe consequences are seen primarily in type 1 (insulin-dependent) diabetes, and death from ketoacidotic coma is rarely seen in people with type 2 (insulin-resistant) diabetes. With insulin treatment, β -oxidation of fatty acids and therefore ketoacid production are reduced, and urinary excretion is able to eliminate the excess acid.

Review Questions

CHAPTER 16: OVERVIEW, GLOMERULAR FILTRATION, AND RENAL CLEARANCE

1. If the clearance of a freely filtered substance is less than the clearance of inulin, then the substance:

- A. is filtered and completely reabsorbed.
- B. is filtered and completely secreted.
- C. underwent net reabsorption.
- D. underwent net secretion.
- E. was neither secreted nor reabsorbed.

2. Diabetic nephropathy is associated with thickened glomerular capillary basement membranes. The decrease in glomerular filtration rate results from:

- A. a reduction in renal blood flow.
- B. a reduction in glomerular capillary hydrostatic pressure.
- C. a reduction in the permeability of the glomerular filtration barrier.
- D. an increase in the glomerular capillary hydrostatic pressure.
- E. an increase in the permeability of the glomerular filtration barrier.

3. The kidneys perform all the following functions EXCEPT:

- A. production of aldosterone.
- B. excretion of excess body acid.
- C. regulation of fluid and electrolyte homeostasis.
- D. activation of vitamin D.
- E. ammoniogenesis.

4. Renal blood flow (RBF) in a young man was determined to be 1 L/min. If his inulin clearance is 125 mL/min, and hematocrit is 0.4, what is his renal filtration fraction?

- A. 15%
- B. 20%
- C. 25%
- D. 30%
- E. 35%

CHAPTER 17: RENAL TRANSPORT PROCESSES

5. Reabsorption of glucose occurs in the:

- A. proximal tubule.
- B. thin descending limb of Henle.

- C. thick ascending limb of Henle.
- D. distal tubule.
- E. collecting duct.

6. Potassium handling by the kidneys is affected by all of the following EXCEPT:

- A. dietary potassium.
- B. plasma potassium concentration.
- C. plasma aldosterone.
- D. plasma antidiuretic hormone.
- E. acidosis.

7. "Loop" diuretics such as furosemide or bumetanide produce a diuresis by targeting which of the following transporters?

- A. Na⁺-glucose cotransporters
- B. Na⁺/H⁺ antiporters
- C. Na⁺/K⁺ ATPase
- D. Na⁺-K⁺-2Cl⁻ cotransporters
- E. Na⁺-HCO₃⁻ cotransporters

CHAPTER 18: URINE CONCENTRATION AND DILUTION MECHANISMS

8. Antidiuretic hormone (ADH)-sensitive water channels (aquaporins) are located in the:

- A. proximal tubule.
- B. thin descending limb of Henle.
- C. thick ascending limb of Henle.
- D. distal tubule.
- E. collecting duct.

9. All of the following factors contribute to establishing or maintaining the medullary interstitial concentration gradient EXCEPT:

- A. distal tubule sodium reabsorption.
- B. Na⁺-K⁺-2Cl⁻ cotransporters on the thick ascending limb of Henle.
- C. solute-free water reabsorption in the descending limb of Henle.
- D. the counter-current multiplier effect.
- E. urea recycling.

10. A patient with chronic pyelonephritis has the following laboratory values:

Urine flow = 3 mL/min $P_{\text{osm}} = 300 \text{ mosm/L}$

$U_{\text{osm}} = 200 \text{ mosm/L}$

The free water clearance in this person is:

- A. -3.
- B. -1.
- C. 0.
- D. +1.
- E. +3.

CHAPTER 19: REGULATION OF EXTRACELLULAR FLUID VOLUME AND OSMOLARITY

11. Select the TRUE statement regarding the renin-angiotensin-aldosterone system.

- A. Renin secretion is stimulated by high tubular fluid sodium concentration.
- B. Angiotensin I is converted to angiotensin II exclusively in the pulmonary circulation.
- C. Angiotensin II directly stimulates proximal tubular sodium reabsorption.
- D. High tubular fluid flow rate in the distal tubule will stimulate renin secretion.
- E. Aldosterone stimulates proximal tubule sodium reabsorption.

12. A severely dehydrated patient is admitted to the Emergency Department. Which of the following would NOT be elevated in this patient?

- A. Plasma atrial natriuretic peptide (ANP)
- B. Plasma renin
- C. Plasma antidiuretic hormone (ADH)
- D. Sympathetic nervous system activity
- E. Plasma aldosterone

13. Diabetes insipidus:

- A. usually involves reduction of renal V2 receptors for antidiuretic hormone (ADH).
- B. is associated with increased water channels (aquaporins) in the collecting ducts.
- C. results in excretion of large quantities of hypertonic urine.
- D. occurs mainly in children.
- E. results in excretion of large quantities of hypotonic urine.

CHAPTER 20: REGULATION OF ACID-BASE BALANCE BY THE KIDNEYS

14. Renal tubular hydrogen secretion *increases*:

- A. when ECF volume is increased.
- B. when arterial PCO_2 is decreased.
- C. when plasma bicarbonate concentration is low due to metabolic acidosis.
- D. when plasma aldosterone is low.
- E. when plasma antidiuretic hormone is low.

15. Net acid excretion (NAE) is dependent on all of the following factors EXCEPT:

- A. excretion of ammonium.
- B. excretion of bicarbonate.
- C. excretion of sodium.
- D. excretion of titratable acids.
- E. distal H^+ secretion.

16. Using the following blood values, determine the acid-base disorder:

pH	Pco_2	HCO_3^-	Na^+	K^+	Cl^-
7.28	26 mm Hg	14 mEq/L	136 mEq/L	5.0 mEq/L	100 mEq/L

- A. Metabolic alkalosis
- B. Metabolic acidosis from acid gain
- C. Metabolic acidosis from diarrhea
- D. Respiratory acidosis
- E. Respiratory alkalosis

17. When compared to normal values, in *acute uncompensated respiratory alkalosis*, plasma:

- A. pH and PCO_2 are high, and HCO_3^- is normal.
- B. pH and PCO_2 are low, and HCO_3^- is high.
- C. pH is high, and PCO_2 and HCO_3^- are low.
- D. pH is high, PCO_2 is low, and HCO_3^- is normal.
- E. pH is high, PCO_2 is low, and HCO_3^- is high.

18. The major titratable acids (TA) excreted in the urine are in the form of:

- A. H_2PO_4^- .
- B. lactic acid.
- C. K_2PO_4 .
- D. H_2CO_3 .
- E. keto acid.

Section 6

GASTROINTESTINAL PHYSIOLOGY

What makes the gastrointestinal (GI) tract interesting? It is an area of our bodies that we are acutely aware of multiple times a day. We have hunger pangs and feel “full”; we become thirsty, hear intestinal sounds, expel gasses, and feel the urge to defecate. Also, GI-related distress and pathology is high on the list of personal and medical complaints. Homeostasis in the whole organism relies on the input of nutrients (gas exchange, food, and water), so the GI tract is a critical component of the supply side.

Chapter 21 **Overview of the Gastrointestinal Tract**

Chapter 22 **Motility through the Gastrointestinal Tract**

Chapter 23 **Gastrointestinal Secretions**

Chapter 24 **Hepatobiliary Function**

Chapter 25 **Digestion and Absorption**

Review Questions

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CHAPTER 21

Overview of the Gastrointestinal Tract

STRUCTURE AND OVERALL FUNCTION OF THE GASTROINTESTINAL TRACT

The GI tract can be thought of as one long tube with an input (mouth) and output (anus), with specialized areas, and direct input from associated organs (liver, pancreas, and gallbladder) (Fig. 21.1). Along the tract there are sphincters that separate major sections of the tract, allowing the regulation of flow of food into the stomach and **chyme** (food and digestive juices) out of the stomach and through the intestines.

A brief tour down the tract:

- **Mouth:** Mechanical digestion occurs through **mastication** (chewing), and chemical digestion begins on the predominant dietary carbohydrate, **starch**, and to a small degree on **lipids**. The salivary glands secrete saliva, which helps buffer, digest, and lubricate the food so it can be swallowed.
- **Esophagus:** As soon as swallowing occurs, the bolus of food enters the esophagus through the relaxation of the upper esophageal sphincter. The esophagus is a tube that allows passage of the bolus of food through the chest and into the stomach. At the end of the esophagus the **lower esophageal sphincter (LES)** relaxes to allow the bolus into the stomach.
- **Stomach:** This is a pouch in the tract that is specialized for the storage of food, secretion of digestive enzymes and hydrochloric acid (HCl), and mixing of the food and digestive juices to make **chyme**. The **pyloric sphincter**, at the distal end of the stomach, regulates the entry of chyme into the small intestine.
- **Small intestine:** The small intestine is made of three sections, the **duodenum** (~1 foot long), the **jejunum** (~8 feet long), and the **ileum** (~10 feet long). Secretions from the liver, gallbladder, and pancreas enter the small intestine via the common bile duct and through the sphincter of Oddi, into the duodenum. A key aspect of the small intestine is the presence of finger-like projections in the intestinal mucosa (the **villus brush border**), which increases the surface area for absorption. Through the tract, there is only one layer of intestinal epithelial

cells, or **enterocytes**, between the lumen of the tract and the systemic blood supply. While this effectively keeps the luminal contents in the small intestine, when absorption occurs, there is rapid and efficient movement of nutrients into the blood. However, the enterocytes are vulnerable to the harsh environment of the gut lumen and need to be protected. This protection is conferred by intestinal buffers and mucus. Despite the protection, intestinal cells will slough off and be replaced by new cells every 3 to 4 days.

The majority of digestion occurs in the jejunum by pancreatic enzymatic action. When nutrients are digested to the constitutive elements (e.g., monosaccharides, monoglycerides, amino acids, etc.), absorption can occur all along the small intestine, although most nutrients are absorbed by the mid- jejunum. The **terminal part of the ileum** is also the site of vitamin B12 absorption and recycling of bile. Chyme that remains in the tract leaves the small intestine through the **ileocecocolic sphincter**, into the large intestine (colon).

- **Large intestine:** The large intestine does not have a brush border, and the primary functions are dehydration of chyme to produce feces, and then storage of the feces until defecation. The large intestine consists of the cecum, appendix, colon, rectum, and anal canal. At the end of the rectum, the **internal and external anal sphincters** regulate expulsion of feces.

Another important aspect of the tract is that its musculature is smooth muscle, except for the mouth, upper esophagus, and external anal sphincter, which have skeletal muscle. Skeletal muscle allows voluntary control of both input (chewing, swallowing) and output (defecating). The rest of the tract has **longitudinal and circular** bands of smooth muscle that allow for propulsion and mixing of the chyme. Figure 21.2 illustrates these muscle layers, as well as the **oblique** layer that is unique to the stomach. The oblique layer aids in the mixing component that is important in stomach motility.

Other distinct aspects of the GI tract include a large blood and lymphatic system, secretion of substances from a variety of glands, and a nervous system that is unique to the GI tract (Fig. 21.3).